

7. The Radon-Nikodym theorem.

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7.1 Densities and conditioning

By Theorem 4.24, any non-negative integrable f defines a Borel measure via:

$$\nu(A) = \int_A f \, dm. \quad (7.1)$$

This chapter establishes precisely which measures admit such a density. It also generalizes the conditional expectation relation, originally derived using joint densities for $A \in \mathcal{F}_X$:

$$\int_A \mathbb{E}(Y|X) \, dP = \int_A Y \, dP, \quad (7.2)$$

to construct $\mathbb{E}(Y|\mathcal{G})$ for any sub- σ -field $\mathcal{G} \subseteq \mathcal{F}$, strictly without assuming the existence of a joint density.

7.2 The Radon-Nikodym theorem

Equation (7.1) implies that if $m(A) = 0$, then $\nu(A) = 0$. This motivates a formal definition.

Definition 7.1. Let $(\Omega, \mathcal{F})$ be a measurable space with measures ν and μ . We say ν is *absolutely continuous with respect to* μ (written $\nu \ll \mu$) if $\mu(A) = 0 \implies \nu(A) = 0$ for all $A \in \mathcal{F}$.

Proposition 7.2. Let ν and μ be finite measures on $(\Omega, \mathcal{F})$. Then $\nu \ll \mu \iff \forall \epsilon > 0, \exists \delta > 0$ such that for $F \in \mathcal{F}$, $\mu(F) < \delta \implies \nu(F) < \epsilon$.

CruX. ($\implies$) Suppose $\nu \ll \mu$ but the condition fails. $\exists \epsilon > 0$ such that for $\delta = 1/2^n$, we can pick $F_n \in \mathcal{F}$ where $\mu(F_n) < 1/2^n$ but $\nu(F_n) \geq \epsilon$. Define $E_n = \bigcup_{i \geq n} F_i$ and the decreasing intersection $A = \bigcap_{n \geq 1} E_n$.

$$\mu(A) \leq \mu(E_n) \leq \sum_{i \geq n} \frac{1}{2^i} = \frac{1}{2^{n-1}}$$

As $n \rightarrow \infty$, $\mu(A) = 0$. Since ν is finite, continuity from above (Theorem 2.19) applies:

$$\nu(A) = \lim_{n \rightarrow \infty} \nu(E_n)$$

Because $\nu(E_n) \geq \nu(F_n) \geq \epsilon$, we have $\nu(A) \geq \epsilon > 0$, contradicting $\nu \ll \mu$.

($\Leftarrow$) If the condition holds and $\mu(F) = 0$, then $\mu(F) < \delta$ for every $\delta > 0$. Thus, $\nu(F) < \epsilon$ for all $\epsilon > 0$, forcing $\nu(F) = 0$. $\square$

Definition 7.3. A measure μ *dominates* ν on $(\Omega, \mathcal{F})$ if for all $A \in \mathcal{F}$: $0 \leq \nu(A) \leq \mu(A)$.

Definition 7.4. A finite (measurable) *partition* $\mathcal{P}$ of Ω is a finite disjoint collection of $\mathcal{F}$ -measurable sets whose union is equal to Ω . The finite partition $\mathcal{P}'$ *refines* $\mathcal{P}$ if every set in $\mathcal{P}$ is a disjoint union of sets in $\mathcal{P}'$. The coarsest common refinement of finite partitions $\mathcal{P}_1$ and $\mathcal{P}_2$ is $\mathcal{P} \triangleq \{A \cap B : A \in \mathcal{P}_1, B \in \mathcal{P}_2\}$.

Theorem 7.5. Suppose that $\mu(\Omega) = 1$ and μ dominates ν . Then $\exists h : \Omega \rightarrow \mathbb{R}$, $h \geq 0$ which is $\mathcal{F}$ -measurable, such that $\forall F \in \mathcal{F}$, $\nu(F) = \int_F h \, d\mu$.

CruX. For any partition $\mathcal{P} = \{A_i\}$, define $h_{\mathcal{P}} = \sum c_i \mathbb{I}_{A_i}$, where $c_i = \nu(A_i)/\mu(A_i)$ (and $c_i = 0$ if $\mu(A_i) = 0$). Since $\nu \leq \mu$, $0 \leq h_{\mathcal{P}} \leq 1$. By linearity, $\nu(A) = \int_A h_{\mathcal{P}} \, d\mu$ for any $A \in \mathcal{P}$.

If $\mathcal{P}_2$ refines $\mathcal{P}_1$ (denoting $h_k = h_{\mathcal{P}_k}$), integrating over any $A \in \mathcal{P}_1$ yields:

- (i) $\int_A h_1 \, d\mu = \nu(A) = \int_A h_2 \, d\mu$.
- (ii) Since h_1 is constant on A , $\int_A h_1(h_2 - h_1) \, d\mu = 0$, implying $\int_A h_2^2 \, d\mu = \int_A h_1^2 \, d\mu + \int_A (h_2 - h_1)^2 \, d\mu$.
- (iii) Summing over all $A \in \mathcal{P}_1$ gives $\int_{\Omega} h_2^2 \, d\mu \geq \int_{\Omega} h_1^2 \, d\mu$.

Let $c \triangleq \sup_{\mathcal{P}} \int_{\Omega} h_{\mathcal{P}}^2 \, d\mu \leq 1$. Choose partitions $\mathcal{P}_n$ such that $\int_{\Omega} h_{\mathcal{P}_n}^2 \, d\mu > c - 4^{-n}$. Let $\mathcal{Q}_n$ be the common refinement of $\mathcal{P}_1, \dots, \mathcal{P}_n$. Since $\mathcal{Q}_{n+1}$ refines $\mathcal{Q}_n$ and $\int_{\Omega} h_{\mathcal{Q}_n}^2 \, d\mu \geq \int_{\Omega} h_{\mathcal{P}_n}^2 \, d\mu > c - 4^{-n}$, property (ii) forces:

$$\int_{\Omega} (h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n})^2 \, d\mu = \int_{\Omega} h_{\mathcal{Q}_{n+1}}^2 \, d\mu - \int_{\Omega} h_{\mathcal{Q}_n}^2 \, d\mu < 4^{-n}.$$

By Cauchy-Schwarz (Theorem 5.5), $\int_{\Omega} |h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}| \, d\mu \leq 2^{-n}$. Beppo-Levi (Theorem 4.30) ensures $\sum |h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}| < \infty$ μ -a.e., guaranteeing the absolute convergence of the telescoping sum. Thus, $h = \lim_n h_{\mathcal{Q}_n}$ exists μ -a.e. (set $h = 0$ on the null set).

For any $F \in \mathcal{F}$, let $\mathcal{R}_n$ refine both $\mathcal{Q}_n$ and $\{F, F^c\}$. The same bounding logic gives $\int_F |h_{\mathcal{R}_n} - h_{\mathcal{Q}_n}| \, d\mu < 2^{-n}$. By property (i), $\nu(F) = \int_F h_{\mathcal{R}_n} \, d\mu$. Splitting this integral and taking $n \rightarrow \infty$:

$$\nu(F) = \lim_{n \rightarrow \infty} \left[\int_F (h_{\mathcal{R}_n} - h_{\mathcal{Q}_n}) \, d\mu + \int_F h_{\mathcal{Q}_n} \, d\mu \right] = 0 + \int_F h \, d\mu$$

where DCT permits passing the limit into the second integral because $h_{\mathcal{Q}_n} \rightarrow h$ pointwise μ -a.e., $0 \leq h_{\mathcal{Q}_n} \leq 1$, and $\mu(\Omega) = 1$. $\square$

Proposition 7.6. *If μ and φ are finite measures with $0 \leq \mu \leq \varphi$, and if $h_\mu = \frac{d\mu}{d\varphi}$ is constructed as above, then for any non-negative $\mathcal{F}$ -measurable function g on Ω we have*

$$\int_{\Omega} g \, d\mu = \int_{\Omega} g h_\mu \, d\varphi.$$

The same identity holds for any $g \in L^1(\mu)$.

CruX. For $g = \mathbb{I}_F$, the definition of h_μ gives $\int_{\Omega} \mathbb{I}_F \, d\mu = \mu(F) = \int_F h_\mu \, d\varphi = \int_{\Omega} \mathbb{I}_F h_\mu \, d\varphi$. By linearity, this holds for any non-negative simple function s . For measurable $g \geq 0$, choose simple functions $s_n \uparrow g$. Since $h_\mu \geq 0$, $s_n h_\mu \uparrow g h_\mu$. The Monotone Convergence Theorem (Theorem 4.13) gives:

$$\int_{\Omega} g \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} s_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} s_n h_\mu \, d\varphi = \int_{\Omega} g h_\mu \, d\varphi.$$

For $g \in L^1(\mu)$, decompose $g = g^+ - g^-$. Both parts have finite integrals, so applying the identity to each and subtracting yields the final result. $\square$

Theorem 7.7. *Let ν and μ be finite measures on the measurable space $(\Omega, \mathcal{F})$ and suppose that $\nu \ll \mu$. Then there is a non-negative $\mathcal{F}$ -measurable function h on Ω such that $\nu(A) = \int_A h \, d\mu$ for all $A \in \mathcal{F}$.*

CruX. Let $\varphi \triangleq \nu + \mu$. Since φ is a finite measure dominating both, Theorem 7.5 provides densities $h_\nu = \frac{d\nu}{d\varphi}$ and $h_\mu = \frac{d\mu}{d\varphi}$. Let $F = \{\omega : h_\mu(\omega) > 0\}$. Then $\mu(F^c) = \int_{F^c} h_\mu \, d\varphi = 0$. Because $\nu \ll \mu$, F^c is also a ν -null set. Thus, for any $A \in \mathcal{F}$, we only need to integrate over $A \cap F$.

Define $h = h_\nu/h_\mu$ on F , and $h = 0$ on F^c . Applying Proposition 7.6 with the test function $g = h \mathbb{I}_{A \cap F}$ yields:

$$\nu(A) = \nu(A \cap F) = \int_{A \cap F} h_\nu \, d\varphi = \int_{A \cap F} h h_\mu \, d\varphi = \int_{A \cap F} h \, d\mu = \int_A h \, d\mu.$$

$\square$

Proposition 7.8 (Radon-Nikodym Theorem). *Given two σ -finite measures ν, μ on a measurable space $(\Omega, \mathcal{F})$, with $\nu \ll \mu$, then there is a non-negative, μ -a.e. finite valued measurable function $h : \Omega \rightarrow \mathbb{R}^+$ such that $\nu(F) = \int_F h \, d\mu$ for every $F \in \mathcal{F}$. The function h is unique up to μ -null sets; if $g : \Omega \rightarrow \mathbb{R}^+$ also satisfies $\nu(F) = \int_F g \, d\mu$ for all $F \in \mathcal{F}$, then $g = h$ μ -a.e.*

CruX. Since ν and μ are σ -finite, partition Ω into disjoint sets $(C_n)_{n=1}^\infty$ where both $\mu(C_n) < \infty$ and $\nu(C_n) < \infty$.

On each $(C_n, \mathcal{F}_n)$, Theorem 7.7 provides a non-negative, $\mathcal{F}$ -measurable h_n (extended to 0 outside C_n) such that $\nu(F \cap C_n) = \int_F h_n \, d\mu$. Because $\nu(C_n) < \infty$, h_n is finite μ -a.e.

Define $h = \sum_{n=1}^{\infty} h_n$. Since the sets C_n are disjoint, at most one term is non-zero for any given ω , guaranteeing h inherits the strictly μ -a.e. finite-valued nature of h_n . For any $F \in \mathcal{F}$:

$$\nu(F) = \sum_{n=1}^{\infty} \nu(F \cap C_n) = \sum_{n=1}^{\infty} \int_F h_n d\mu = \int_F h d\mu$$

where the final equality holds by applying the Monotone Convergence Theorem (Theorem 4.13) to the partial sums of h_n .

Uniqueness: If g also satisfies $\nu(F) = \int_F g d\mu$, then for any $F \in \mathcal{F}$, $\int_{F \cap C_n} h d\mu = \nu(F \cap C_n) = \int_{F \cap C_n} g d\mu$. Since both are integrable on C_n , Proposition 4.22(ii) forces $h = g$ μ -a.e. on C_n . The countable union of these μ -null sets across all C_n remains a μ -null set, proving $h = g$ μ -a.e. on Ω . $\square$

Corollary 7.8.33. *If ν and μ are σ -finite measures with $\nu \ll \mu$, and if $h_\nu = \frac{d\nu}{d\mu}$ is constructed as in Theorem 7.8, then for any non-negative $\mathcal{F}$ -measurable function g on Ω , we have*

$$\int_{\Omega} g d\nu = \int_{\Omega} g h_\nu d\mu.$$

The same identity holds for any $g \in L^1(\nu)$.

CruX. For $g = \mathbb{I}_F$, the Radon-Nikodym Theorem directly yields $\int_{\Omega} \mathbb{I}_F d\nu = \nu(F) = \int_F h_\nu d\mu = \int_{\Omega} \mathbb{I}_F h_\nu d\mu$. By linearity, this holds for all non-negative simple functions g_n .

For measurable $g \geq 0$, take simple $g_n \uparrow g$. Since $h_\nu \geq 0$, $g_n h_\nu \uparrow g h_\nu$. The Monotone Convergence Theorem yields:

$$\int_{\Omega} g d\nu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\nu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n h_\nu d\mu = \int_{\Omega} g h_\nu d\mu.$$

For $g \in L^1(\nu)$, split $g = g^+ - g^-$. Both parts are non-negative and have finite integrals, so the identity holds for g by linearity. $\square$

Note: By replacing g with the restricted function $g \mathbb{I}_A$ for any $A \in \mathcal{F}$, the identities in Proposition 7.6 and Corollary 7.8.33 immediately localize to the set A :

$$\int_A g d\nu = \int_A g h_\nu d\mu.$$

Lemma 7.8.66. *Let μ be a measure on $(\Omega, \mathcal{F})$, and let h, g be non-negative $\mathcal{F}$ -measurable functions. If the measure defined by $\rho(F) = \int_F h d\mu$ is σ -finite, and if $\int_F h d\mu = \int_F g d\mu$ for all $F \in \mathcal{F}$, then $h = g$ μ -a.e.*

The conclusion holds when $h, g \in L^1(\mu)$ instead of being restricted to non-negative functions.

CruX. Since ρ is σ -finite, partition Ω into disjoint sets $(C_n)_{n=1}^\infty$ where $\rho(C_n) = \int_{C_n} h \, d\mu < \infty$. Equality of integrals guarantees $\int_{C_n} g \, d\mu < \infty$, making both functions integrable on each C_n . Because $\int_{F \cap C_n} h \, d\mu = \int_{F \cap C_n} g \, d\mu$ for all $F \in \mathcal{F}$, Proposition 4.22(ii) forces $h = g$ μ -a.e. on C_n . Taking the countable union of these null sets proves $h = g$ μ -a.e. on Ω .

For $h, g \in L^1(\mu)$, decompose them: $h = h^+ - h^-$ and $g = g^+ - g^-$. The hypothesis is $\int_F h^+ - h^- \, d\mu = \int_F g^+ - g^- \, d\mu$. Because all parts are in $L^1(\mu)$, their integrals are finite, allowing us to rearrange terms without $\infty - \infty$ conflicts:

$$\int_F (h^+ + g^-) \, d\mu = \int_F (g^+ + h^-) \, d\mu.$$

Both $H \triangleq h^+ + g^-$ and $G \triangleq g^+ + h^-$ are non-negative and integrable. Applying the non-negative result yields $H = G$ μ -a.e. Since L^1 functions are finite almost everywhere, we subtract h^- and g^- pointwise to conclude $h^+ - h^- = g^+ - g^-$ μ -a.e., which is exactly $h = g$ μ -a.e. $\square$

Proposition 7.9. Assume we are given σ -finite measures λ, ν, μ satisfying $\lambda \ll \mu$ and $\nu \ll \mu$ with Radon-Nikodym derivatives $\frac{d\lambda}{d\mu}$ and $\frac{d\nu}{d\mu}$, respectively.

- (i) With $\phi = \lambda + \nu$, we have $\frac{d\phi}{d\mu} = \frac{d\lambda}{d\mu} + \frac{d\nu}{d\mu}$ μ -a.e.
- (ii) If, in addition, $\lambda \ll \nu$, then $\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu} \frac{d\nu}{d\mu}$ μ -a.e.

CruX. (i) Since $\lambda \ll \mu$ and $\nu \ll \mu$, their sum satisfies $\phi \ll \mu$. By the Radon-Nikodym Theorem (Theorem 7.8) and linearity:

$$\int_F \frac{d\phi}{d\mu} \, d\mu = \phi(F) = \lambda(F) + \nu(F) = \int_F \frac{d\lambda}{d\mu} \, d\mu + \int_F \frac{d\nu}{d\mu} \, d\mu = \int_F \left(\frac{d\lambda}{d\mu} + \frac{d\nu}{d\mu} \right) \, d\mu.$$

Since λ and ν are σ -finite, ϕ is σ -finite. Applying Lemma 7.8.66 to these equivalent integrals yields $\frac{d\phi}{d\mu} = \frac{d\lambda}{d\mu} + \frac{d\nu}{d\mu}$ μ -a.e.

- (ii) For any $F \in \mathcal{F}$, Theorem 7.8 gives $\lambda(F) = \int_F \frac{d\lambda}{d\mu} \, d\mu$ and $\lambda(F) = \int_F \frac{d\lambda}{d\nu} \, d\nu$. Applying Corollary 7.8.33 to the second integral using $g = \frac{d\lambda}{d\nu} \mathbb{I}_F$ translates the measure from ν to μ :

$$\int_F \frac{d\lambda}{d\nu} \, d\nu = \int_F \frac{d\lambda}{d\nu} \frac{d\nu}{d\mu} \, d\mu.$$

Equating the two μ -integrals yields $\int_F \frac{d\lambda}{d\mu} \, d\mu = \int_F \frac{d\lambda}{d\nu} \frac{d\nu}{d\mu} \, d\mu$. Since λ is σ -finite, Lemma 7.8.66 immediately forces $\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu} \frac{d\nu}{d\mu}$ μ -a.e. $\square$

Proposition 7.9.5. Let ν and μ be two σ -finite measures that are equivalent (i.e., both $\nu \ll \mu$ and $\mu \ll \nu$). Then

$$\frac{d\mu}{d\nu} = \left(\frac{d\nu}{d\mu} \right)^{-1} \quad \mu\text{-a.e.}$$

Cru. Let $g = \frac{d\nu}{d\mu}$ and $h = \frac{d\mu}{d\nu}$. By definition of the measure μ , the Radon-Nikodym Theorem, and Corollary 7.8.33:

$$\mu(F) = \int_F 1 \, d\mu = \int_F h \, d\nu = \int_F hg \, d\mu.$$

Because μ is given as σ -finite, Lemma 7.8.66 immediately implies $gh = 1$ μ -a.e.

To ensure the inverse is well-defined, let $N = \{\omega : h(\omega) = 0\}$ and $M = \{\omega : g(\omega) = 0\}$. Since $h = 0$ on N , $\mu(N) = \int_N h \, d\nu = 0$. Similarly, since $g = 0$ on M , $\nu(M) = \int_M g \, d\mu = 0$. Because $\mu \ll \nu$, this implies $\mu(M) = 0$ as well. Therefore, the set $M \cup N$ where either function is zero is a μ -null set, allowing us to safely invert g almost everywhere:

$$\frac{d\mu}{d\nu} = \left(\frac{d\nu}{d\mu} \right)^{-1} \quad \mu\text{-a.e.}$$

□

Definition 7.10. If there is a set $E \in \mathcal{F}$ such that $\lambda(F) = \lambda(E \cap F)$ for every $F \in \mathcal{F}$, then λ is said to be *concentrated on E* . If two measures μ and ν are concentrated on disjoint subsets of Ω , we say that they are *mutually singular* and write $\mu \perp \nu$.

Lemma 7.10.5. *A measure λ is concentrated on a set $E \in \mathcal{F}$ if and only if $\forall F \in \mathcal{F}$, the condition $F \cap E = \emptyset$ implies $\lambda(F) = 0$.*

Cru. ($\implies$) If λ is concentrated on E and $E \cap F = \emptyset$, then $\lambda(F) = \lambda(E \cap F) = \lambda(\emptyset) = 0$.

($\impliedby$) Assume that $F' \cap E = \emptyset \implies \lambda(F') = 0$. For any $F \in \mathcal{F}$, partition it as $F = (F \cap E) \cup (F \setminus E)$. Because $(F \setminus E) \cap E = \emptyset$, the hypothesis gives $\lambda(F \setminus E) = 0$. Additivity then yields $\lambda(F) = \lambda(E \cap F) + 0$, meaning λ is concentrated on E . □

Proposition 7.11. *If μ, ν, λ_1 and λ_2 are measures on a σ -field $\mathcal{F}$, the following are true:*

- (i) *If $\lambda_1 \perp \mu$ and $\lambda_2 \perp \mu$, then $\lambda_1 + \lambda_2 \perp \mu$.*
- (ii) *If $\lambda_1 \ll \mu$ and $\lambda_2 \perp \mu$, then $\lambda_2 \perp \lambda_1$.*
- (iii) *If $\nu \ll \mu$ and $\nu \perp \mu$, then $\nu = 0$.*

Cru. (i) By hypothesis, there exist disjoint pairs (A_1, B_1) and (A_2, B_2) such that λ_i is concentrated on A_i and μ is concentrated on B_i . For any $F \in \mathcal{F}$, additivity gives $(\lambda_1 + \lambda_2)(F) = \lambda_1(F \cap A_1) + \lambda_2(F \cap A_2)$. Because λ_1 and λ_2 are concentrated on A_1 and A_2 respectively, this is exactly equal to $(\lambda_1 + \lambda_2)(F \cap (A_1 \cup A_2))$, proving the sum is concentrated on $A_1 \cup A_2$. For μ , being concentrated on both B_1 and B_2 means $\mu(F) = \mu(F \cap B_1) = \mu(F \cap B_1 \cap B_2)$, so μ is concentrated on $B_1 \cap B_2$. Since $(A_1 \cup A_2) \cap (B_1 \cap B_2) \subseteq (A_1 \cap B_1) \cup (A_2 \cap B_2) = \emptyset$, they are mutually singular.

- (ii) Suppose μ is concentrated on E . By Lemma 7.10.5, any $F \in \mathcal{F}$ with $F \cap E = \emptyset$ has $\mu(F) = 0$. Because $\lambda_1 \ll \mu$, this implies $\lambda_1(F) = 0$, meaning λ_1 is also concentrated on E . By definition of $\lambda_2 \perp \mu$, λ_2 is concentrated on E^c . Thus, $\lambda_2 \perp \lambda_1$.

- (iii) Applying case (ii) with $\lambda_1 = \lambda_2 = \nu$ yields $\nu \perp \nu$. This means ν must be concentrated on some set A and its disjoint complement A^c . Thus, for the entire space, $\nu(\Omega) = \nu(\Omega \cap A \cap A^c) = \nu(\emptyset) = 0$. $\square$

Lemma 7.11.5. *Let μ and λ be measures on a measurable space $(\Omega, \mathcal{F})$, and define the measure $\varphi = \mu + \lambda$. For any non-negative $\mathcal{F}$ -measurable function h , or any $h \in L^1(\varphi)$, and any set $E \in \mathcal{F}$, we have*

$$\int_E h \, d\varphi = \int_E h \, d\mu + \int_E h \, d\lambda.$$

Cruz. For an indicator function $h = \mathbb{I}_A$, the definition of the measure φ yields:

$$\int_E \mathbb{I}_A \, d\varphi = \varphi(A \cap E) = \mu(A \cap E) + \lambda(A \cap E) = \int_E \mathbb{I}_A \, d\mu + \int_E \mathbb{I}_A \, d\lambda.$$

By linearity, this identity strictly extends to any non-negative simple function s .

For any non-negative $\mathcal{F}$ -measurable function h , take simple functions $s_n \uparrow h$. The equality holds for s_n , and invoking the Monotone Convergence Theorem (Theorem 4.13) pushes the limit inside the integrals:

$$\int_E h \, d\varphi = \lim_{n \rightarrow \infty} \int_E s_n \, d\varphi = \lim_{n \rightarrow \infty} \left(\int_E s_n \, d\mu + \int_E s_n \, d\lambda \right) = \int_E h \, d\mu + \int_E h \, d\lambda.$$

For $h \in L^1(\varphi)$, decompose $h = h^+ - h^-$. Because h is φ -integrable, $\int_E h^+ \, d\varphi$ and $\int_E h^- \, d\varphi$ are finite, forcing the respective μ and λ integrals to also be finite. Applying the result to h^+ and h^- separately and subtracting yields the result by linearity. $\square$

Theorem 7.12 (Lebesgue Decomposition). *Let λ, μ be σ -finite measures on $(\Omega, \mathcal{F})$. Then λ can be expressed uniquely as a sum of two measures, $\lambda = \lambda_a + \lambda_s$ where $\lambda_a \ll \mu$ and $\lambda_s \perp \mu$.*

Cruz. Finite Case: Assume λ, μ are finite. Let $\varphi = \lambda + \mu$. Since φ dominates λ , Theorem 7.5 yields an h ($0 \leq h \leq 1$) such that $\lambda(F) = \int_F h \, d\varphi$. By Lemma 7.11.5, $\lambda(F) = \int_F h \, d\lambda + \int_F h \, d\mu$. Let $A = \{h < 1\}$ and $B = \{h = 1\}$. Define $\lambda_a(F) \triangleq \lambda(F \cap A)$ and $\lambda_s(F) \triangleq \lambda(F \cap B)$, so $\lambda = \lambda_a + \lambda_s$.

If $\mu(N) = 0$, then $\lambda(N \cap A) = \int_{N \cap A} h \, d\lambda + \int_{N \cap A} h \, d\mu = \int_{N \cap A} h \, d\lambda$, implying $\int_{N \cap A} (1 - h) \, d\lambda = 0$. Since $1 - h > 0$ on A , we must have $\lambda(N \cap A) = \lambda_a(N) = 0$ (see the note after Proposition 4.17). Thus, $\lambda_a \ll \mu$. On $F \cap B$, $h = 1$. So $\lambda(F \cap B) = \int_{F \cap B} 1 \, d\lambda + \int_{F \cap B} 1 \, d\mu = \lambda(F \cap B) + \mu(F \cap B)$, forcing $\mu(F \cap B) = 0$. Since μ is concentrated on A and λ_s on B , $\lambda_s \perp \mu$.

σ -finite Case: Partition Ω into disjoint sets (C_n) where both measures are finite. Applying the finite case to each $(C_n, \mathcal{F}_n)$ yields disjoint partitions $A_n \cup B_n = C_n$. Define

$A = \bigcup A_n$ and $B = \bigcup B_n$. Because the C_n are disjoint, $A \cap B = \emptyset$ and $A \cup B = \Omega$. Countable additivity gives $\lambda(F) = \sum \lambda(F \cap C_n) = \lambda(F \cap A) + \lambda(F \cap B) = \lambda_a(F) + \lambda_s(F)$. If $\mu(N) = 0$, then $\mu(N \cap C_n) = 0 \implies \lambda(N \cap A_n) = 0$, so $\lambda_a(N) = \lambda(N \cap A) = 0$. Similarly, $\mu(F \cap B_n) = 0 \implies \mu(F \cap B) = 0$, proving $\lambda_a \ll \mu$ and $\lambda_s \perp \mu$.

Uniqueness: Suppose $\lambda = \lambda_{a'} + \lambda_{s'}$ with $\lambda_{a'} \ll \mu$ and $\lambda_{s'} \perp \mu$. Let $\lambda_s, \lambda_{s'}$ be concentrated on B, B' respectively. Let $E = B \cup B'$. Since $\mu(B) = \mu(B') = 0$, $\mu(E) = 0$. Absolute continuity forces $\lambda_a(E) = \lambda_{a'}(E) = 0$. For any $F \in \mathcal{F}$, partition $F = (F \cap E) \cup (F \setminus E)$. On $F \cap E$:

$$\lambda(F \cap E) = \lambda_a(F \cap E) + \lambda_s(F \cap E) = \lambda_s(F \cap E).$$

Similarly, $\lambda(F \cap E) = \lambda_{s'}(F \cap E)$. For $F \setminus E \subseteq E^c = B^c \cap (B')^c$, both singular measures vanish: $\lambda_s(F \setminus E) = \lambda_{s'}(F \setminus E) = 0$. Adding the pieces yields $\lambda_s = \lambda_{s'}$.

On any C_n , $\lambda_a(F \cap C_n) + \lambda_s(F \cap C_n) = \lambda_{a'}(F \cap C_n) + \lambda_{s'}(F \cap C_n)$. Since $\lambda_s = \lambda_{s'}$ and $\lambda_s(F \cap C_n) \leq \lambda(C_n) < \infty$, we subtract the finite singular terms to get $\lambda_a(F \cap C_n) = \lambda_{a'}(F \cap C_n)$. Summing over n proves $\lambda_a = \lambda_{a'}$. $\square$

Corollary 7.13 (L-R-N Decomposition). *With μ , λ , λ_a and λ_s as in the previous theorem, there is a μ -a.s. unique non-negative measurable function h such that*

$$\lambda(E) = \int_E h \, d\mu + \lambda_s(E)$$

for every $E \in \mathcal{F}$.

CruX. Since $\lambda_a \ll \mu$, the Radon-Nikodym Theorem (Theorem 7.8) guarantees a μ -a.s. unique non-negative measurable h such that $\lambda_a(E) = \int_E h \, d\mu$. Substituting this directly into the Lebesgue Decomposition $\lambda(E) = \lambda_a(E) + \lambda_s(E)$ yields the result immediately. $\square$

7.3 Lebesgue-Stieltjes measures

Proposition 7.15. *If $F : \mathbb{R} \rightarrow \mathbb{R}$ is monotone increasing, then the left-limit $F(x-)$ and the right-limit $F(x+)$ exist at every $x \in \mathbb{R}$ and $F(x-) \leq F(x) \leq F(x+)$. Hence F has at most countably many discontinuities, and these are jump discontinuities, i.e. $F(x-) < F(x+)$.*

CruX. Fix $x_0 \in \mathbb{R}$. By monotonicity, the sets $\{F(x) : x < x_0\}$ and $\{F(x) : x > x_0\}$ are bounded above and below by $F(x_0)$, respectively. Thus, $l = \sup_{x < x_0} F(x)$ and $r = \inf_{x > x_0} F(x)$ exist, and $l \leq F(x_0) \leq r$.

For any $\epsilon > 0$, the definition of the supremum guarantees an $x' < x_0$ such that $l - \epsilon < F(x') \leq l$. By monotonicity, $l - \epsilon < F(x) \leq l$ for all $x \in (x', x_0)$. This proves $F(x_0-) = l$. Symmetrical logic for the infimum proves $F(x_0+) = r$. Thus, $F(x_0-) \leq F(x_0) \leq F(x_0+)$. A discontinuity occurs if and only if $F(x_0-) < F(x_0+)$, which is strictly a jump discontinuity.

For every jump discontinuity at x , choose a rational number $q_x \in (F(x-), F(x+))$. If $x_1 < x_2$, monotonicity forces $F(x_1+) \leq F(x_2-)$, meaning the jump intervals are strictly disjoint. The mapping $x \mapsto q_x$ is therefore an injection from the set of discontinuities into the countable set $\mathbb{Q}$, proving the discontinuities are at most countable. $\square$

Definition 7.16. For a monotonically increasing right-continuous function F , we define *length relative to F* of a bounded interval $(a, b]$ as $l_F((a, b]) \triangleq F(b) - F(a)$. The *F -outer measure* of any set $A \subseteq \mathbb{R}$ is the element of $[0, \infty]$ defined as

$$m_F^*(A) \triangleq \inf Z_F(A)$$

where

$$Z_F(A) \triangleq \left\{ \sum_{n=1}^{\infty} l_F(I_n) : I_n = (a_n, b_n], a_n \leq b_n, A \subseteq \bigcup_{n=1}^{\infty} I_n \right\}.$$

Theorem 2.6 (revisited). *The F -outer measure of the half-open interval $I = (a, b]$ equals its length relative to F i.e.*

$$m_F^*(I) = l_F(I).$$

CruX. (i) $m_F^*(I) \leq l_F(I)$: Let the sequence of intervals (I_n) be such that $I_1 = I = (a, b]$ and $I_n = \emptyset$ for $n > 1$. Then $\sum_{n=1}^{\infty} l_F(I_n) = l_F(I) \in Z_F(I)$ as (I_n) covers I . Since the outer measure is the infimum over all such covers, $m_F^*(I) = \inf Z_F(I) \leq l_F(I)$.
(ii) $l_F(I) \leq m_F^*(I)$: Let $\epsilon > 0$. By the definition of the infimum, there exists a countable cover $I_n = (a_n, b_n]$ such that $\sum_{n=1}^{\infty} l_F(I_n) \leq m_F^*(I) + \frac{\epsilon}{2}$. Since F is right-continuous, for each n , there exists $b'_n > b_n$ such that $F(b'_n) - F(b_n) < \frac{\epsilon}{2^{n+1}}$. Define the open intervals $J_n = (a_n, b'_n)$. Thus, $F(b'_n) - F(a_n) < l_F(I_n) + \frac{\epsilon}{2^{n+1}}$. Fix $0 < \delta < b - a$. The sequence (J_n) is an open cover of the compact interval $[a + \delta, b]$. By the Heine-Borel theorem, a finite subcover $(J_n)_{n=1}^N$ exists. Assume WLOG that this finite subcover is minimally ordered, meaning $a_1 < a + \delta < b'_1$, and $a_k < b'_{k-1} < b'_k$ for $k = 2, \dots, N$, with $b \leq b'_N$. Because F is monotonically increasing, $a_k < b'_{k-1} \implies F(a_k) \leq F(b'_{k-1})$. This allows us to chain the lengths:

$$\begin{aligned} F(b) - F(a + \delta) &\leq F(b'_N) - F(a_1) \\ &= F(b'_N) - F(a_N) + F(a_N) - F(a_{N-1}) + \dots - F(a_1) \\ &\leq F(b'_N) - F(a_N) + F(b'_{N-1}) - F(a_{N-1}) + \dots - F(a_1) \\ &\leq \sum_{n=1}^N (F(b'_n) - F(a_n)). \end{aligned}$$

Stringing everything together yields:

$$\begin{aligned}
m_F^*(I) + \epsilon &\geq \sum_{n=1}^{\infty} l_F(I_n) + \frac{\epsilon}{2} \\
&\geq \sum_{n=1}^N \left(F(b_n) - F(a_n) + \frac{\epsilon}{2^{n+1}} \right) \\
&> \sum_{n=1}^N (F(b'_n) - F(a_n)) \\
&\geq F(b) - F(a + \delta).
\end{aligned}$$

Since $\epsilon > 0$ is arbitrary, $F(b) - F(a + \delta) \leq m_F^*(I)$. Finally, because F is right-continuous, taking the limit as $\delta \rightarrow 0^+$ gives $F(b) - F(a) = l_F(I) \leq m_F^*(I)$. $\square$

Note: m_F^* satisfies

- (i) **Non-negativity:** $m_F^*(A) \geq 0, \forall A \subseteq \mathbb{R}, m_F^*(\emptyset) = 0$,
- (ii) **Monotonicity:** $m_F^*(B) \leq m_F^*(A)$ if $B \subseteq A$, and
- (iii) **Countable Sub-additivity:** $m_F^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} m_F^*(A_n)$.

However, **translational invariance (Proposition 2.8) does not hold in general for m_F^* .**

Definition 2.9 (for m_F^*). A set $E \subseteq \mathbb{R}$ is measurable (for the outer measure m_F^*) if for every set $A \subseteq \mathbb{R}$ we have

$$m_F^*(A) = m_F^*(A \cap E) + m_F^*(A \cap E^c)$$

where $E^c = \mathbb{R} \setminus E$ and we write $E \in \mathcal{M}_F$.

The restriction of m_F^* to the σ -field $\mathcal{M}_F$ of *Lebesgue-Stieltjes measurable sets*, is denoted by m_F .

Summary: The Lebesgue-Stieltjes Extension (Theorems 2.10 – 2.31)

The topological and set-theoretic machinery developed for the Lebesgue outer measure m^* applies verbatim to the Lebesgue-Stieltjes outer measure m_F^* . Therefore, all structural results from Theorem 2.10 through Proposition 2.31 hold for m_F^* and its associated Carathéodory σ -field $\mathcal{M}_F$, with the sole exception of translation invariance (Proposition 2.15). Where standard proofs require expanding an interval's length by ϵ (e.g., Theorem 2.17), the right-continuity of F guarantees a spatial stretch $\delta > 0$ such that $F(b + \delta) - F(b) < \epsilon$.

Crucially, because the Borel σ -field $\mathcal{B}$ is generated by the standard topology of open intervals in $\mathbb{R}$, it is completely independent of the underlying measure. Thus, $\mathcal{B}_F = \mathcal{B}$.

Following the exact logic of Theorem 2.28, the σ -field $\mathcal{M}_F$ is precisely the completion of the Borel σ -field $\mathcal{B}$ with respect to the measure m_F .

Finally, a singleton $\{a\}$ need not be a null set. If F is discontinuous at a , then $F(a) > F(a-)$, resulting in strictly positive measure: $m_F(\{a\}) = F(a) - F(a-) > 0$.

Definition 7.20. A real function F is *absolutely continuous* (AC) on the interval $[a, b]$ if, given $\epsilon > 0$, there is $\delta > 0$ such that for every finite set of disjoint intervals $J_k = (x_k, y_k)$, $k = 1, \dots, n$ contained in $[a, b]$ with $\sum_{k=1}^n (y_k - x_k) < \delta$, we have $\sum_{k=1}^n |F(y_k) - F(x_k)| < \epsilon$.

Proposition 7.21. If $f \in L^1([a, b])$ for a finite interval $[a, b]$, then the function $F(x) \triangleq \int_{[a, x]} f \, dm$ is absolutely continuous on $[a, b]$.

CruX. Given $\epsilon > 0$, define the measure $\mu(A) \triangleq \int_A |f| \, dm$. Because $f \in L^1([a, b])$, μ is a finite measure. By the definition of the Lebesgue integral, $m(A) = 0 \implies \mu(A) = 0$, guaranteeing $\mu \ll m$. Since both μ and m are finite measures, Proposition 7.2 ensures $\exists \delta > 0$ such that $m(A) < \delta \implies \mu(A) < \epsilon$.

Consider any finite set of disjoint intervals $J_k = (x_k, y_k)$ in $[a, b]$ satisfying $\sum_{k=1}^n (y_k - x_k) < \delta$. Let $E \triangleq \bigcup_{k=1}^n J_k$. Because $m(E) < \delta$, we strictly have $\mu(E) < \epsilon$. Evaluating the function differences over these intervals yields:

$$\begin{aligned} \sum_{k=1}^n |F(y_k) - F(x_k)| &= \sum_{k=1}^n \left| \int_{J_k} f \, dm \right| \\ &\leq \sum_{k=1}^n \int_{J_k} |f| \, dm \\ &= \mu(E) < \epsilon. \end{aligned}$$

Thus, F perfectly satisfies the (ϵ, δ) definition of absolute continuity. $\square$

Theorem 7.22. Let F be a monotone increasing and absolutely continuous function on $\mathbb{R}$, and let m_F be the Lebesgue-Stieltjes measure it generates. Then every Lebesgue-measurable set is m_F -measurable, and on these sets, $m_F \ll m$.

CruX. Since F is absolutely continuous, given $\epsilon > 0$, choose $\delta > 0$ such that for any finite set of disjoint open intervals $I_k = (x_k, y_k)$, $\sum_{k=1}^n m(I_k) < \delta \implies \sum_{k=1}^n |F(y_k) - F(x_k)| < \epsilon/2$.

Let $B \in \mathcal{B}$ be a Borel null set ($m(B) = 0$). By Theorem 2.17, there exists an open set $O \supset B$ such that $m(O) < \delta$. Express O as a countable union of disjoint open intervals $I_k = (x_k, y_k)$. Since $\sum_{k=1}^n m(I_k) \leq m(O) < \delta$ for any integer n , absolute continuity forces $\sum_{k=1}^n |F(y_k) - F(x_k)| < \epsilon/2$. Taking $n \rightarrow \infty$ yields $\sum_{k=1}^{\infty} |F(y_k) - F(x_k)| \leq \epsilon/2 < \epsilon$. Because $m_F(O) = \sum_{k=1}^{\infty} |F(y_k) - F(x_k)|$ and $B \subseteq O$, we have $m_F(B) \leq m_F(O) < \epsilon$. Since $\epsilon > 0$ is arbitrary, $m_F(B) = 0$.

Any Lebesgue null set $N \in \mathcal{M}$ is contained in a Borel null set B_N (Theorem 2.17(ii)). By the previous step, $m_F(B_N) = 0$, forcing $m_F(N) = 0$. Thus, m -null sets are m_F -null sets ($m_F \ll m$) for monotone AC functions F .

For any Lebesgue measurable set $E \in \mathcal{M}$, there exists a Borel set $B \supseteq E$ such that $m(B \setminus E) = 0$. Since $B \setminus E$ is an m -null set, it is an m_F -null set, making it m_F -measurable. Since $E = B \setminus (B \setminus E)$, and σ -algebras are closed under relative complements, we strictly conclude $E \in \mathcal{M}_F$ so that $\mathcal{M} \subseteq \mathcal{M}_F$. $\square$

Definition 7.23. A real function F is of *bounded variation* on $[a, b]$ (briefly $F \in BV[a, b]$) if the *total variation function* of F , namely $T_F[a, b] < \infty$, where for any $x \in [a, b]$,

$$T_F[a, x] \triangleq \sup \left\{ \sum_{k=1}^n |F(x_k) - F(x_{k-1})| \right\}$$

with the supremum taken over all finite partitions of $[a, x]$ with $a = x_0 < x_1 < x_2 \cdots < x_n = x$.

Similarly the *positive variation function* $P_F[a, x]$ and the *negative variation function* $N_F[a, x]$ of F are defined as

$$P_F[a, x] \triangleq \sup \left\{ \sum_{k=1}^n [F(x_k) - F(x_{k-1})]^+ \right\}$$

$$N_F[a, x] \triangleq \sup \left\{ \sum_{k=1}^n [F(x_k) - F(x_{k-1})]^- \right\}$$

where the supremum is again taken over all finite partitions of $[a, x]$. Fixing a , we treat these as functions of $x \geq a$, denoted as $P_F(x)$, $N_F(x)$ and $T_F(x)$.

Proposition 7.24. If F is of bounded variation on $[a, b]$, we have $F(x) - F(a) = P_F(x) - N_F(x)$, while $T_F(x) = P_F(x) + N_F(x)$ for $x \in [a, b]$.

CruX. For any finite partition $\mathcal{P}$ of $[a, x]$, let the variation sums be $n(\mathcal{P})$, $p(\mathcal{P})$, and $t(\mathcal{P})$. A refinement $\mathcal{P}_1 \cup \mathcal{P}_2$ strictly increases or maintains the sums. For any partition, we strictly have the identities $t(\mathcal{P}) = p(\mathcal{P}) + n(\mathcal{P})$ and $F(x) - F(a) = p(\mathcal{P}) - n(\mathcal{P})$.

Second Identity: $p(\mathcal{P}_1) + n(\mathcal{P}_2) \leq t(\mathcal{P}_1 \cup \mathcal{P}_2) \leq T_F(x)$. Taking the suprema over $\mathcal{P}_1$ and $\mathcal{P}_2$ independently yields $P_F(x) + N_F(x) \leq T_F(x)$. Conversely, $t(\mathcal{P}) = p(\mathcal{P}) + n(\mathcal{P}) \leq P_F(x) + N_F(x)$. Taking the supremum over $\mathcal{P}$ yields $T_F(x) \leq P_F(x) + N_F(x)$, proving $T_F(x) = P_F(x) + N_F(x)$.

First Identity: Using $p(\mathcal{P}_1 \cup \mathcal{P}_2) - n(\mathcal{P}_1 \cup \mathcal{P}_2) = F(x) - F(a)$: Since $-N_F(x) \leq -n(\mathcal{P}_1 \cup \mathcal{P}_2)$, we have $p(\mathcal{P}_1) - N_F(x) \leq F(x) - F(a)$. Taking the supremum over $\mathcal{P}_1$ yields $P_F(x) - N_F(x) \leq F(x) - F(a)$. Similarly, since $-P_F(x) \leq -p(\mathcal{P}_1 \cup \mathcal{P}_2)$, we have $n(\mathcal{P}_2) - P_F(x) \leq F(a) - F(x)$. Taking the supremum over $\mathcal{P}_2$ gives $N_F(x) - P_F(x) \leq F(a) - F(x)$, which rearranges to $F(x) - F(a) \leq P_F(x) - N_F(x)$. Thus, $F(x) - F(a) = P_F(x) - N_F(x)$. $\square$

Proposition 7.25. *If F is of bounded variation and $a \leq x \leq b$ then $T_F[a, b] = T_F[a, x] + T_F[x, b]$. Similar results hold for P_F and N_F . Hence all three variation functions are monotone increasing in x for fixed $a \in \mathbb{R}$. Moreover, if F has bounded variation on $[a, b]$, then it has bounded variation on any $[c, d] \subset [a, b]$.*

CruX. Let $\mathcal{P}$ be any finite partition of $[a, b]$. The refinement $\mathcal{Q} = \mathcal{P} \cup \{x\}$ splits into partitions $\mathcal{Q}_1$ of $[a, x]$ and $\mathcal{Q}_2$ of $[x, b]$. Since refinement increases variation, $t(\mathcal{P}) \leq t(\mathcal{Q}_1) + t(\mathcal{Q}_2) \leq T_F[a, x] + T_F[x, b]$. Taking the supremum over $\mathcal{P}$ yields $T_F[a, b] \leq T_F[a, x] + T_F[x, b]$.

Conversely, for any partitions $\mathcal{Q}_1$ of $[a, x]$ and $\mathcal{Q}_2$ of $[x, b]$, their union $\mathcal{Q}$ partitions $[a, b]$. Thus, $t(\mathcal{Q}_1) + t(\mathcal{Q}_2) = t(\mathcal{Q}) \leq T_F[a, b]$. Taking independent suprema over $\mathcal{Q}_1$ and $\mathcal{Q}_2$ yields $T_F[a, x] + T_F[x, b] \leq T_F[a, b]$. Equality holds, and the logic is identical for P_F and N_F .

Because variations are non-negative, $T_F[a, b] = T_F[a, x] + T_F[x, b] \implies T_F[a, x] \leq T_F[a, b]$. Thus, the variation functions are monotonically increasing.

Finally, if $[c, d] \subset [a, b]$, applying the identity gives $T_F[a, b] = T_F[a, c] + T_F[c, d] + T_F[d, b]$. Since $F \in BV[a, b]$, $T_F[a, b] < \infty$, strictly forcing the non-negative component $T_F[c, d] < \infty$. Hence, $F \in BV[c, d]$. $\square$

Theorem 7.26. *Let $[a, b]$ be a finite interval. A real function is of bounded variation on $[a, b]$ if and only if it is the difference of two monotone increasing real functions which are in $BV[a, b]$.*

CruX. • ($\implies$) If $F \in BV[a, b]$, Proposition 7.24 guarantees $F(x) - F(a) = P_F(x) - N_F(x)$. Rearranging yields $F(x) = (F(a) + P_F(x)) - N_F(x)$. By Proposition 7.25, $P_F(x)$ and $N_F(x)$ are monotonic increasing, making F the difference of two monotone increasing real functions. Since the total variation functions of $F(a) + P_F(x)$ and $N_F(x)$ on $[a, b]$ are equal to $P_F(b) - P_F(a)$ and $N_F(b) - N_F(a)$ respectively (constant and finite), both functions are in $BV[a, b]$.

• ($\impliedby$) Suppose $F = F_1 - F_2$ where F_1, F_2 are monotonic increasing on $[a, b]$. For any finite partition of $[a, b]$, the triangle inequality yields:

$$\begin{aligned} \sum_{k=1}^n |F(x_k) - F(x_{k-1})| &= \sum_{k=1}^n |(F_1(x_k) - F_1(x_{k-1})) - (F_2(x_k) - F_2(x_{k-1}))| \\ &\leq \sum_{k=1}^n |F_1(x_k) - F_1(x_{k-1})| + \sum_{k=1}^n |F_2(x_k) - F_2(x_{k-1})|. \end{aligned}$$

Because F_1, F_2 are monotonic increasing, the absolute values drop. The sums perfectly telescope to:

$$(F_1(b) - F_1(a)) + (F_2(b) - F_2(a)).$$

Since F_1 and F_2 are real-valued, this sum is strictly finite and independent of the partition. Taking the supremum over all partitions confirms $F \in BV[a, b]$. $\square$

Exercise 7.13

- (a) If F is monotone increasing on $[a, b]$, find $T_F[a, b]$.
- (b) If $F \in BV[a, b]$, then F is continuous m -a.e. and hence is Lebesgue measurable.
- (c) A differentiable function on $[0, 1]$ need not be in $BV[0, 1]$.
- (d) If F is Lipschitz-continuous on $[a, b]$ ($\exists M$ such that $0 < M < \infty$, and for all $x, y \in [a, b]$, $|F(x) - F(y)| \leq M|x - y|$), then $F \in BV[a, b]$.

CruX. (a) Using the respective definitions, it is easy to show that $T_F[a, b] = P_F[a, b] = F(b) - F(a)$, and $N_F[a, b] = 0$.

- (b) By Theorem 7.26, $F(x) = (F(a) + P_F(x)) - N_F(x)$, the difference of two monotone increasing functions. By Proposition 7.15, monotone functions have at most countably many discontinuities. Thus, F has at most countably many discontinuities, making it continuous m -a.e. Since m -a.e. continuous functions are Lebesgue measurable, F is Lebesgue measurable.
- (c) Consider $F(x) = x^2 \sin(x^{-2})$ for $x > 0$ with $F(0) = 0$. The derivative exists everywhere in $[0, 1]$ (with $F'(0) = 0$). However, evaluating $F(x)$ at the local extrema $x_n = 1/\sqrt{n\pi + \pi/2}$ yields variations proportional to the harmonic series $\sum \frac{1}{n} = \infty$. Thus, $F \notin BV[0, 1]$.
- (d) For any partition, $\sum |F(x_k) - F(x_{k-1})| \leq \sum M|x_k - x_{k-1}| = M(b - a)$. Taking the supremum gives $T_F[a, b] \leq M(b - a) < \infty$, hence $F \in BV[a, b]$.

□

Proposition 7.27. Suppose the real function F is absolutely continuous on $[a, b]$; then we have:

- (i) $F \in BV[a, b]$,
- (ii) If $F = F_1 - F_2$ is the minimal decomposition of F as the difference of two monotone increasing functions described in Theorem 7.26, then both F_1 and F_2 are absolutely continuous on $[a, b]$.

CruX. (i) Given $\epsilon = 1$, absolute continuity provides a $\delta > 0$. Divide $[a, b]$ into a fixed number $K = \lceil (b - a)/\delta \rceil$ of subintervals $I_j = [t_{j-1}, t_j]$ each of length less than δ . For any arbitrary finite partition $\mathcal{P}$ of $[a, b]$, the refinement $\mathcal{P} \cup \{t_0, \dots, t_K\}$ splits the variation sum across these K blocks. Within each block, the total interval length is $< \delta$, so the variation sum within that block is strictly < 1 . Thus, $t(\mathcal{P}) < K \cdot 1 < \infty$ for all $\mathcal{P}$, proving $T_F[a, b] \leq K < \infty$, so $F \in BV[a, b]$.

- (ii) Let $\epsilon > 0$ be given, and let $\delta > 0$ be the corresponding parameter from the absolute continuity condition of F for $\epsilon/2$. Consider any finite collection of disjoint intervals $(a_k, b_k) \subset [a, b]$ with $\sum_{k=1}^m (b_k - a_k) < \delta$.

The definition of the supremum allows us to choose a finite partition $\mathcal{P}_k$ within each $[a_k, b_k]$ such that $P_F[a_k, b_k] \leq \sum_{i \in \mathcal{P}_k} [F(x_i) - F(x_{i-1})]^+ + \frac{\epsilon}{2m}$. Collecting all sub-intervals from all $\mathcal{P}_k$ forms a combined set of disjoint intervals whose aggregate length is strictly less than δ . By the absolute continuity of F :

$$\begin{aligned} \sum_{k=1}^m |P_F(b_k) - P_F(a_k)| &= \sum_{k=1}^m P_F[a_k, b_k] \leq \sum_{k=1}^m \sum_{i \in \mathcal{P}_k} [F(x_i) - F(x_{i-1})]^+ + \epsilon/2 \\ &\leq \sum_{k=1}^m \sum_{i \in \mathcal{P}_k} |F(x_i) - F(x_{i-1})| + \epsilon/2 \\ &< \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

Thus, $\sum_{k=1}^m |P_F(b_k) - P_F(a_k)| < \epsilon$, proving P_F (and thus F_1) is absolutely continuous. Since $F_2(x) = N_F(x) = P_F(x) - F(x) + F(a)$, F_2 is the difference of absolutely continuous functions, making it absolutely continuous as well. $\square$

Note on Minimality: Let $F = G_1 - G_2$ be any other valid decomposition of F into monotone increasing functions. For any $x < y$, $F(y) - F(x) = (G_1(y) - G_1(x)) - (G_2(y) - G_2(x))$. Since G_2 is increasing, its difference is non-negative, strictly forcing:

$$F(y) - F(x) \leq G_1(y) - G_1(x)$$

Because G_1 is also increasing, its difference is non-negative, allowing us to safely take the positive part:

$$[F(y) - F(x)]^+ \leq G_1(y) - G_1(x)$$

Summing this inequality over any partition of $[a, x]$ and taking the supremum yields $P_F(x) - P_F(a) \leq G_1(x) - G_1(a)$. Thus, P_F strictly lower-bounds the growth of any other positive component G_1 . By symmetry, N_F lower-bounds any G_2 . Therefore, $F_1 = P_F + F(a)$ and $F_2 = N_F$ represent the absolute minimum growth required to reconstruct F , forming the minimal decomposition of F . This is unique up to a simultaneous additive constant.

Definition 7.28. If $F \in BV[a, b]$, let $F = F_1 - F_2$ be its minimal decomposition into monotone increasing functions. The Lebesgue-Stieltjes *signed measure* determined by F is the countably additive set function m_F defined on the Borel σ -field $\mathcal{B}$ by $m_F = m_{F_1} - m_{F_2}$, where m_{F_i} are the Lebesgue-Stieltjes measures determined by F_i .

Note on the Domain of m_F : While Definition 7.28 restricts the domain of m_F to $\mathcal{B}$ for a general $F \in BV[a, b]$, it naturally extends to the entire Lebesgue σ -algebra $\mathcal{M}$ when F is absolutely continuous. For an AC function, its monotonic components F_1, F_2 are also AC, ensuring $m_{F_1} \ll m$ and $m_{F_2} \ll m$. Consequently, any m -null set is a null set for both m_{F_1} and m_{F_2} . Thus, the completion of $\mathcal{B}$ with respect to m is fully contained in the completions with respect to m_{F_i} , strictly meaning $\mathcal{M} \subseteq \mathcal{M}_{F_1} \cap \mathcal{M}_{F_2}$.

Note on BV vs. AC: Proposition 7.27 proves $AC \implies BV$, but the converse is strictly false. The classic counterexample is the Cantor ternary function (the “Devil’s Staircase”), which is continuous and monotonic (hence BV) but fails to be AC as its derivative is 0 m -a.e. and hence, cannot be recovered by integrating its derivative contradicting Theorem 7.36.

The Devil’s Staircase is the quintessential *singular* function. A continuous function F is *singular* if its induced Lebesgue-Stieltjes measure is mutually singular to the Lebesgue measure ($m_F \perp m$), concentrating all its growth on a Lebesgue null set.

The structural gap between BV and AC is formalized by the following decomposition stated here without proof:

Theorem 7.28.5 (Lebesgue Decomposition of Functions). *Any function $F \in BV[a, b]$ can be uniquely decomposed (up to additive constants) as:*

$$F = F_{AC} + F_{Jump} + F_{Singular}$$

where F_{AC} is an absolutely continuous function, F_{Jump} is a pure jump function constructed from the countable discontinuities of F , and $F_{Singular}$ is a continuous singular function ($m_{F_{Singular}} \perp m$, or equivalently $F'_{Singular} = 0$ m -a.e.).

Definition 7.30. A *signed measure* on a measurable space $(\Omega, \mathcal{F})$ is a set function $\nu : \mathcal{F} \rightarrow (-\infty, \infty]$ satisfying:

- (i) $\nu(\emptyset) = 0$
- (ii) $\nu(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \nu(E_i)$ if $E_i \in \mathcal{F}$ and $E_i \cap E_j = \emptyset$ for $i \neq j$.

Since the LHS in (ii) is invariant under rearrangements of the sets,

$$\left| \nu \left(\bigcup_{i=1}^{\infty} E_i \right) \right| < \infty \iff \sum_{i=1}^{\infty} |\nu(E_i)| < \infty.$$

Also, if $E, F \in \mathcal{F}$, $F \subseteq E$, and $|\nu(E)| < \infty$, then because $\nu(E \setminus F) > -\infty$ and $\nu(E) = \nu(F) + \nu(E \setminus F)$ is finite, we must strictly have $|\nu(F)| < \infty$.

Definition 7.30.33. Let ν and μ be measures (either standard positive measures or signed measures) on a measurable space $(\Omega, \mathcal{F})$.

A set $E \in \mathcal{F}$ is called μ -null if for every set $A \in \mathcal{F}$ such that $A \subseteq E$, we have $\mu(A) = 0$.

The measure ν is *absolutely continuous* with respect to μ , denoted by $\nu \ll \mu$, if every μ -null set is also a ν -null set.

Note: This universally covers all combinations of positive and signed measures. If a measure is positive, monotonicity guarantees that $\mu(E) = 0$ is sufficient to force $\mu(A) = 0$ for all $A \subseteq E$, perfectly reducing to the classical definition of absolute continuity.

Proposition 7.30.66. *If F is an absolutely continuous function on $[a, b]$, then its Lebesgue-Stieltjes signed measure m_F is absolutely continuous with respect to the Lebesgue measure m ($m_F \ll m$).*

Cru. While Definition 7.28 defines m_F on the Borel sets, the absolute continuity of F guarantees m_F naturally extends to every Lebesgue measurable set. Thus, the relation $m_F \ll m$ is well-defined over the entire Lebesgue σ -algebra $\mathcal{M}$.

By Proposition 7.27, F admits a minimal decomposition $F = F_1 - F_2$ into monotonically increasing, absolutely continuous functions on $[a, b]$. The associated signed measure is $m_F = m_{F_1} - m_{F_2}$.

Because F_1, F_2 are monotonic and absolutely continuous, Theorem 7.22 ensures their induced positive measures satisfy $m_{F_1} \ll m$ and $m_{F_2} \ll m$.

Let E be an m -null set, and let $A \subseteq E$ be any measurable subset. Since m is positive, monotonicity dictates $m(A) \leq m(E) = 0$. The absolute continuity of the positive components forces $m_{F_1}(A) = 0$ and $m_{F_2}(A) = 0$. Therefore,

$$m_F(A) = m_{F_1}(A) - m_{F_2}(A) = 0.$$

Since $m_F(A) = 0$ for every measurable subset $A \subseteq E$, E is an m_F -null set (Definition 7.30.33), proving $m_F \ll m$. $\square$

Proposition 7.31. *A signed measure ν defined on a σ -field $\mathcal{F}$ is monotone increasing ($F \subseteq E \implies \nu(F) \leq \nu(E)$) if and only if ν is a measure on $\mathcal{F}$.*

Cru. If ν is a measure, it is inherently monotone increasing, making the forward direction trivial. Conversely, if ν is a monotone increasing signed measure, then because $\emptyset \subseteq E$ for all $E \in \mathcal{F}$, we strictly have $\nu(E) \geq \nu(\emptyset) = 0$. Since ν is non-negative and countably additive, it is a standard positive measure. $\square$

Definition 7.31.5. Let $\nu : \mathcal{F} \rightarrow \mathbb{R}$ be a bounded signed measure. The *total variation* of ν , denoted by $|\nu|$, is a set function on $\mathcal{F}$ defined by

$$|\nu|(E) \triangleq \sup \left\{ \sum_{i=1}^{\infty} |\nu(E_i)| : \{E_i\}_{i=1}^{\infty} \subset \mathcal{F}, E = \bigcup_{i=1}^{\infty} E_i, E_i \cap E_j = \emptyset \text{ if } i \neq j \right\},$$

from which it is immediate that $\forall E \in \mathcal{F}, |\nu(E)| \leq |\nu|(E)$. Hence $|\nu|$ dominates ν .

Theorem 7.32. *The total variation $|\nu|$ of a bounded signed measure ν is a (positive) finite measure on $\mathcal{F}$.*

Cru. Let (E_i) be a countable partition of $E \in \mathcal{F}$. We must show $|\nu|(E) = \sum_i |\nu|(E_i)$.

($\geq$) Let $\epsilon > 0$. For each E_i , choose a partition $(A_{ij})_{j=1}^\infty$ of E_i such that $\sum_{j=1}^\infty |\nu(A_{ij})| \geq |\nu|(E_i) - \epsilon/2^i$. Summing over i yields:

$$\sum_{i=1}^\infty \sum_{j=1}^\infty |\nu(A_{ij})| \geq \sum_{i=1}^\infty |\nu|(E_i) - \epsilon.$$

Since $\{A_{ij}\}$ partitions E , its sum is strictly bounded above by $|\nu|(E)$. Thus, $|\nu|(E) \geq \sum_i |\nu|(E_i) - \epsilon$. As $\epsilon > 0$ is arbitrary, $|\nu|(E) \geq \sum_i |\nu|(E_i)$.

($\leq$) Let (B_k) be any partition of E . The intersections $(E_i \cap B_k)$ form a common refinement. By countable additivity and the triangle inequality:

$$\sum_{k=1}^\infty |\nu(B_k)| = \sum_{k=1}^\infty \left| \sum_{i=1}^\infty \nu(E_i \cap B_k) \right| \leq \sum_{k=1}^\infty \sum_{i=1}^\infty |\nu(E_i \cap B_k)| = \sum_{i=1}^\infty \left(\sum_{k=1}^\infty |\nu(E_i \cap B_k)| \right).$$

For a fixed i , $(E_i \cap B_k)_{k \geq 1}$ partitions E_i , so its sum is bounded by $|\nu|(E_i)$. Therefore:

$$\sum_{k=1}^\infty |\nu(B_k)| \leq \sum_{i=1}^\infty |\nu|(E_i).$$

Taking the supremum over all partitions (B_k) yields $|\nu|(E) \leq \sum_i |\nu|(E_i)$.

Combined, $|\nu|$ is countably additive. Non-negativity and $|\nu|(\emptyset) = 0$ are trivial, confirming $|\nu|$ is a positive measure.

Finiteness: Note ν is bounded, so $\exists M < \infty$ where $|\nu(A)| \leq M$ for all $A \in \mathcal{F}$. For any partition $\{E_1, \dots, E_n\}$ of Ω , group the elements into A (where $\nu(E_i) \geq 0$) and B (where $\nu(E_i) < 0$). Then $\sum |\nu(E_i)| = \nu(A) - \nu(B) \leq |\nu(A)| + |\nu(B)| \leq 2M$. Since this holds for any partition, $|\nu|(\Omega) \leq 2M < \infty$. $\square$

Definition 7.32.20. Since $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{F}$, the set functions $\nu^+ \triangleq (|\nu| + \nu)/2$ and $\nu^- \triangleq (|\nu| - \nu)/2$ are countably additive and strictly non-negative. These are the *positive variation* and *negative variation* of ν , acting as standard positive measures on $\mathcal{F}$. They recover the original components:

$$\nu = \nu^+ - \nu^- \quad \text{and} \quad |\nu| = \nu^+ + \nu^-.$$

The decomposition $\nu = \nu^+ - \nu^-$ is the *Jordan Decomposition* of ν .

Note on Properties of Signed Measures: Since a bounded signed measure can be decomposed into two finite positive measures ($\nu = \nu^+ - \nu^-$), properties of standard measures that depend strictly on limits and additivity translate perfectly to signed measures by applying them to ν^+ and ν^- individually:

- **Countable Union of Null Sets:** A countable union of ν -null sets is strictly a ν -null set. Any measurable subset of the union can be disjointified into pieces residing entirely within the original null sets. By definition, these pieces have zero measure, and countable additivity ensures their sum is zero.
- **Continuity from Below:** If $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_{n=1}^{\infty} A_n$, then $\lim_{n \rightarrow \infty} \nu(A_n) = \nu(A)$.
- **Continuity from Above:** If $A_1 \supseteq A_2 \supseteq \dots$ and $A = \bigcap_{n=1}^{\infty} A_n$, then $\lim_{n \rightarrow \infty} \nu(A_n) = \nu(A)$.

Also note that while continuity from above requires boundedness (to avoid $\infty - \infty$), continuity from below holds even for unbounded signed measures.

Beware: Properties relying strictly on *non-negativity* completely fail for signed measures!

- **Monotonicity fails:** If $A \subseteq B$, it is generally **not** true that $\nu(A) \leq \nu(B)$, because $B \setminus A$ might carry a net negative mass.
- **Subadditivity (Boole's Inequality) fails:** $\nu(A \cup B) \leq \nu(A) + \nu(B)$ is no longer reliably true due to potential negative cancellations.

Rule of Thumb: To safely bound the measure of subsets or unions for signed measures, always route the inequality through the total variation measure $|\nu|$, which is strictly positive and monotonic.

Proposition 7.32.40 (Minimality of the Jordan Decomposition). *The Jordan decomposition of a bounded signed measure $\nu = \nu^+ - \nu^-$ is minimal. That is, if $\nu = \lambda_1 - \lambda_2$ is any other decomposition of the signed measure ν into two positive measures, then for every measurable set $E \in \mathcal{F}$, we have $\nu^+(E) \leq \lambda_1(E)$ and $\nu^-(E) \leq \lambda_2(E)$.*

CruX. Let $\nu = \lambda_1 - \lambda_2$ for positive measures λ_1, λ_2 . For any measurable set E and any countable partition $\{E_i\}$ of E , the triangle inequality yields:

$$|\nu(E_i)| = |\lambda_1(E_i) - \lambda_2(E_i)| \leq \lambda_1(E_i) + \lambda_2(E_i)$$

Summing over the partition, we obtain:

$$\sum_{i=1}^{\infty} |\nu(E_i)| \leq \sum_{i=1}^{\infty} (\lambda_1(E_i) + \lambda_2(E_i)) = \lambda_1(E) + \lambda_2(E)$$

Taking the supremum over all such partitions of E establishes the bound on the total variation measure:

$$|\nu|(E) \leq \lambda_1(E) + \lambda_2(E)$$

Applying this strictly to the definitions of the positive and negative variations yields:

$$\begin{aligned}\nu^+(E) &= \frac{1}{2}(|\nu|(E) + \nu(E)) \leq \frac{1}{2}(\lambda_1(E) + \lambda_2(E) + \lambda_1(E) - \lambda_2(E)) = \lambda_1(E) \\ \nu^-(E) &= \frac{1}{2}(|\nu|(E) - \nu(E)) \leq \frac{1}{2}(\lambda_1(E) + \lambda_2(E) - (\lambda_1(E) - \lambda_2(E))) = \lambda_2(E)\end{aligned}$$

Thus, ν^+ and ν^- are strictly dominated by any other positive measure decomposition. $\square$

Definition 7.32.60. Let ν be a signed measure on a measurable space $(\Omega, \mathcal{F})$. We say that ν is *concentrated* on a set $A \in \mathcal{F}$ if for every measurable set $E \in \mathcal{F}$, we have:

$$\nu(E) = \nu(E \cap A)$$

Equivalently, ν is concentrated on A if its complement A^c is a ν -null set (i.e., for every $B \in \mathcal{F}$ such that $B \subseteq A^c$, $\nu(B) = 0$).

Furthermore, two measures μ and ν (either positive or signed) are *mutually singular*, denoted by $\mu \perp \nu$, if there exist disjoint sets $A, B \in \mathcal{F}$ such that μ is concentrated on A and ν is concentrated on B .

Note: For signed measures, the simple condition $\nu(A^c) = 0$ is insufficient to define concentration due to the possibility of positive and negative subsets canceling each other out. Strictly defining concentration via null sets extends the classical mutual singularity definition to signed measures seamlessly.

Lemma 7.32.80 (Extension of 7.11 to signed measures). Let μ, ν, λ_1 , and λ_2 be signed measures on a measurable space $(\Omega, \mathcal{F})$. The following hold true:

- (i) If $\lambda_1 \perp \mu$ and $\lambda_2 \perp \mu$, then $\lambda_1 + \lambda_2 \perp \mu$.
- (ii) If $\lambda_1 \ll \mu$ and $\lambda_2 \perp \mu$, then $\lambda_2 \perp \lambda_1$.
- (iii) If $\nu \ll \mu$ and $\nu \perp \mu$, then $\nu = 0$.

CruX. (i) By hypothesis, μ is concentrated on A_1 and A_2 , while λ_1, λ_2 are concentrated on disjoint complements B_1, B_2 respectively. Let $A = A_1 \cap A_2$ and $B = B_1 \cup B_2$. Since $A_i \cap B_i = \emptyset$, we have $A \cap B = \emptyset$. Because $A^c = A_1^c \cup A_2^c$ is the union of two μ -null sets, A^c is μ -null, meaning μ is concentrated on A . Because $B^c = B_1^c \cap B_2^c \subseteq B_1^c$, B^c is a λ_1 -null set. Similarly, $B^c \subseteq B_2^c$, making it a λ_2 -null set. Thus, B^c is a $(\lambda_1 + \lambda_2)$ -null set. Hence, $\lambda_1 + \lambda_2$ is concentrated on B , proving $(\lambda_1 + \lambda_2) \perp \mu$.

- (ii) $\exists A, B, A \cap B = \emptyset$, such that μ is concentrated on A (A^c is μ -null) and λ_2 is concentrated on B . As $\lambda_1 \ll \mu$, the μ -null set A^c is strictly a λ_1 -null set, meaning λ_1 is concentrated on A . Since λ_1 is concentrated on A and λ_2 is concentrated on B with $A \cap B = \emptyset$, we have $\lambda_2 \perp \lambda_1$.

- (iii) $\exists A, B, A \cap B = \emptyset$, such that μ is concentrated on A (A^c is μ -null) and ν is concentrated on B (B^c is ν -null). Since $A \cap B = \emptyset \implies B \subseteq A^c$, B is a μ -null set. Because $\nu \ll \mu$, B being μ -null strictly forces B to be a ν -null set. Since both B and B^c are ν -null sets, their union $\Omega = B \cup B^c$ is a ν -null set, meaning $\nu = 0$. $\square$

Theorem 7.33. *Let μ be a σ -finite (positive) measure and suppose that ν is a bounded signed measure. Then there is a unique decomposition $\nu = \nu_a + \nu_s$, into two signed measures, with $\nu_a \ll \mu$ and $\nu_s \perp \mu$. Moreover, there is a unique (up to sets of μ -measure 0) $h \in L^1(\mu)$ such that $\nu_a(F) = \int_F h \, d\mu$ for all $F \in \mathcal{F}$.*

CruX. Use the Jordan decomposition to express $\nu = \nu^+ - \nu^-$, where ν^+, ν^- are finite positive measures. Apply the standard (positive) Lebesgue-Radon-Nikodym theorem to both variations individually:

$$\begin{aligned} \nu^+ &= \nu_a^+ + \nu_s^+ \quad \text{with} \quad \nu_a^+ \ll \mu, \quad \nu_s^+ \perp \mu, \quad \text{and} \quad \nu_a^+(F) = \int_F h' \, d\mu \\ \nu^- &= \nu_a^- + \nu_s^- \quad \text{with} \quad \nu_a^- \ll \mu, \quad \nu_s^- \perp \mu, \quad \text{and} \quad \nu_a^-(F) = \int_F h'' \, d\mu \end{aligned}$$

where $h', h'' \in L^1(\mu)$.

Define $\nu_a \triangleq \nu_a^+ - \nu_a^-$ and $\nu_s \triangleq \nu_s^+ - \nu_s^-$, yielding $\nu = \nu_a + \nu_s$. Setting $h \triangleq h' - h'' \in L^1(\mu)$ yields $\nu_a(F) = \int_F h \, d\mu$.

1. **Absolute Continuity:** Let $N \in \mathcal{F}$ be μ -null. For any subset $N' \subseteq N$, monotonicity of μ ensures $\mu(N') = 0$. Since $\nu_a^+, \nu_a^- \ll \mu$, this forces $\nu_a^+(N') = \nu_a^-(N') = 0$, meaning $\nu_a(N') = 0$. Thus, every μ -null set is a ν_a -null set, proving $\nu_a \ll \mu$.
2. **Mutual Singularity:** ν_s^+ and ν_s^- are concentrated on μ -null sets A^+ and A^- . Their union $A = A^+ \cup A^-$ is also μ -null. Both measures are thus concentrated on A , and for any $E \in \mathcal{F}$:

$$\nu_s(E) = \nu_s^+(E \cap A^+) - \nu_s^-(E \cap A^-) = \nu_s^+(E \cap A) - \nu_s^-(E \cap A) = \nu_s(E \cap A).$$

Thus, ν_s is concentrated on the μ -null set A , proving $\nu_s \perp \mu$.

Uniqueness: If $\nu = \lambda_a + \lambda_s$ is another decomposition, then $\nu_a + \nu_s = \lambda_a + \lambda_s \implies \nu_a - \lambda_a = \lambda_s - \nu_s$. The LHS is absolutely continuous with respect to μ , while the RHS is mutually singular to μ . By Lemma 7.32.80(iii), the only measure both absolutely continuous and mutually singular to μ is the zero measure. Hence, $\nu_a = \lambda_a$ and $\nu_s = \lambda_s$. If another function $g \in L^1(\mu)$ exists for ν_a , then $\int_F (h - g) \, d\mu = 0$ for all $F \in \mathcal{F}$, strictly forcing $h = g$ μ -a.e. $\square$

Corollary 7.33.5 (Radon-Nikodym Theorem for signed measures). *Let μ be a σ -finite (positive) measure and suppose that ν is a bounded signed measure such that $\nu \ll \mu$. Then there is a unique (up to sets of μ -measure 0) function $h \in L^1(\mu)$ such that $\nu(E) = \int_E h \, d\mu$ for all $E \in \mathcal{F}$.*

Cru. By Theorem 7.33, ν admits a unique Lebesgue decomposition $\nu = \nu_a + \nu_s$, where $\nu_a \ll \mu$, $\nu_s \perp \mu$, and there exists a unique $h \in L^1(\mu)$ such that $\nu_a(E) = \int_E h \, d\mu$.

Since $\nu \ll \mu$ (given) and $\nu_a \ll \mu$ (by definition), their difference $\nu_s = \nu - \nu_a$ must also be absolutely continuous with respect to μ ($\nu_s \ll \mu$).

However, the decomposition guarantees $\nu_s \perp \mu$. By Lemma 7.32.80(iii), the only signed measure simultaneously absolutely continuous and mutually singular to μ is the zero measure.

Therefore, $\nu_s = 0$. This forces $\nu = \nu_a$, meaning $\nu(E) = \int_E h \, d\mu$. The uniqueness of h follows directly from Theorem 7.33. $\square$

Theorem 7.35. *If ν is a bounded signed measure on $\mathbb{R}$ and $F(x) = \nu((-\infty, x])$, then for any $a \in \mathbb{R}$, the following are equivalent:*

- (i) F is differentiable at a , and $F'(a) = L$.
- (ii) given $\epsilon > 0$ there exists $\delta > 0$ such that $\left| \frac{\nu(J)}{m(J)} - L \right| < \epsilon$ if the open interval J contains a and $m(J) < \delta$.

Cru. Assume WLOG $L = 0$. (If $L \neq 0$, restrict ν to $I = [a - 1, a + 1]$ and consider $\rho = \tilde{\nu} - Lm$. For small intervals $J \subseteq I$, $\frac{\rho(J)}{m(J)} \rightarrow 0 \iff \frac{\nu(J)}{m(J)} \rightarrow L$, and the associated cumulative function $G(x)$ satisfies $G'(a) = 0 \iff F'(a) = L$.)

(i) $\implies$ (ii): Assume $F'(a) = 0$. Given $\epsilon > 0$, choose $\delta > 0$ such that $|z - a| < \delta \implies |F(z) - F(a)| < \frac{\epsilon}{2}|z - a|$. Let $J = (x, y)$ be an open interval containing a with $m(J) < \delta$. Choose $x_k \downarrow x$ and $y_k \uparrow y$ such that $x < x_k < y_k < y$ for all k . For large k , $x < x_k < a \leq y_k < y$, ensuring both x_k and y_k are within δ of a . Routing variation through a :

$$\begin{aligned} |\nu((x_k, y_k))| &= |F(y_k) - F(x_k)| \\ &\leq |F(y_k) - F(a)| + |F(a) - F(x_k)| \\ &< \frac{\epsilon}{2}(y_k - a) + \frac{\epsilon}{2}(a - x_k) = \frac{\epsilon}{2}(y_k - x_k) < \frac{\epsilon}{2}m(J). \end{aligned}$$

Since $\bigcup_{k=1}^{\infty} (x_k, y_k] = (x, y) = J$, continuity from below yields $|\nu(J)| = \lim |\nu((x_k, y_k))| \leq \frac{\epsilon}{2}m(J) < \epsilon m(J)$, proving (ii).

(ii) $\implies$ (i): Let ϵ, δ be as in (ii). For $x < a < y$ with $y - x < \delta$, consider open intervals $J_n = (x, y + 1/n)$. For large n , $m(J_n) < \delta$, so $|\nu(J_n)| < \epsilon(y + 1/n - x)$. Since $J_n \downarrow (x, y]$ and ν is bounded, continuity from above yields:

$$|\nu((x, y])| = \lim_{n \rightarrow \infty} |\nu(J_n)| \leq \epsilon(y - x).$$

This means $|F(y) - F(x)| \leq \epsilon(y - x)$.

To establish continuity at a , let $I_n = (a - 1/n, a + 1/n)$. For large n , $m(I_n) < \delta$, so $|\nu(I_n)| < \epsilon(2/n)$. Since $I_n \downarrow \{a\}$, continuity from above yields $|\nu(\{a\})| = 0$. Since right-continuity is guaranteed, this forces $F(a-) = F(a) = F(a+)$, making F continuous at a .

With continuity established, we can safely take limits in $|F(y) - F(x)| \leq \epsilon(y - x)$. Letting $x \uparrow a$ proves the right-derivative is zero; letting $y \downarrow a$ proves the left-derivative is zero. Thus, $F'(a) = 0$, proving (i). $\square$

Corollary 7.35.33. *Let ν be a finite signed measure on the measurable space $([a, b], \mathcal{M}_{[a, b]})$, and let $F(x) = \nu([a, x])$ for all $x \in [a, b]$. Suppose $x \in (a, b)$ and (J_n) is a sequence of open intervals in $[a, b]$ containing x such that $m(J_n) \rightarrow 0$ as $n \rightarrow \infty$. If*

$$\lim_{n \rightarrow \infty} \frac{\nu(J_n)}{m(J_n)} = L,$$

then F is differentiable at x and $F'(x) = L$.

Proof. Extend ν to a signed measure $\tilde{\nu}$ on the real line by defining $\tilde{\nu}(E) = \nu(E \cap [a, b])$ for any Lebesgue measurable set E . Define $\tilde{F}(x) = \tilde{\nu}((-\infty, x])$. By construction, $\tilde{F}(x) = 0$ for $x < a$, and $\tilde{F}(x) = F(x)$ for $x \in [a, b]$. Because the intervals J_n are eventually contained strictly within the interior (a, b) , the local limit behavior of $\tilde{\nu}$ and $\tilde{F}$ at x is identical to that of ν and F . Applying Theorem 7.35 to $\tilde{F}$ on $\mathbb{R}$ yields the result. $\square$

Note on the Lebesgue Differentiation Theorem: The following theorem bridges the critical gap between abstract measure-theoretic densities (Radon-Nikodym) and geometric derivatives over intervals. The proof requires the Vitali Covering Lemma and is beyond the current scope, but the result is strictly necessary to complete the Fundamental Theorem of Calculus.

Theorem 7.35.66 (Lebesgue Differentiation Theorem). *If $f \in L^1([a, b], m)$, then for m -almost every $x \in [a, b]$,*

$$\lim_{\substack{m(J) \rightarrow 0 \\ x \in J}} \frac{1}{m(J)} \int_J f \, dm = f(x)$$

where the limit is taken over all open intervals $J \subseteq [a, b]$ containing x . Points x where this holds are called Lebesgue points of f .

Theorem 7.36 (Fundamental Theorem of Calculus). *Let F be absolutely continuous on $[a, b]$. Then F is differentiable m -a.e. and its Lebesgue-Stieltjes signed measure m_F has Radon-Nikodym derivative $\frac{dm_F}{dm} = F'$ m -a.e. Moreover, for each $x \in [a, b]$,*

$$F(x) - F(a) = m_F([a, x]) = \int_a^x F'(t) \, dt.$$

Cruæ. Note: The underlying measurable space is strictly $([a, b], \mathcal{M}_{[a, b]})$, and m denotes the Lebesgue measure restricted to $[a, b]$.

Since F is absolutely continuous on $[a, b]$, Proposition 7.30.66 guarantees $m_F \ll m$. Because the domain is bounded, m is finite. The Radon-Nikodym theorem (Proposition 7.8) provides a function $h \in L^1(m)$ such that $m_F(E) = \int_E h \, dm$ for all measurable sets $E \subseteq [a, b]$.

Let $n \in \mathbb{N}$. Define $\delta_n \triangleq (b - a)/2^n$ and $t_i \triangleq a + i \cdot \delta_n$ for $0 \leq i \leq 2^n$. We form the sequence of dyadic partitions $\mathcal{P}_n \triangleq \{(t_i, t_{i+1}] : 0 \leq i < 2^n\}$, where each partition refines the previous one.

Define $h_n(a) = 0$, and for $x \in (a, b]$:

$$h_n(x) \triangleq \sum_{i=0}^{2^n-1} \frac{m_F((t_i, t_{i+1}])}{m((t_i, t_{i+1}])} \mathbb{I}_{(t_i, t_{i+1}]}(x) = \sum_{i=0}^{2^n-1} \frac{F(t_{i+1}) - F(t_i)}{t_{i+1} - t_i} \mathbb{I}_{(t_i, t_{i+1}]}(x).$$

By the Lebesgue Differentiation Theorem (LDT) (Theorem 7.35.66), since $h \in L^1(m)$, almost every point in $[a, b]$ is a Lebesgue point for h . Let D be the countable set of all dyadic endpoints across all n . Let N be the m -null set of points that are not Lebesgue points for h . Fix any $x \in (a, b) \setminus (D \cup N)$.

Because $x \notin D$, for every n , x belongs strictly to the interior of exactly one partition interval. Let $J_n = (t_i, t_{i+1})$ be this open interval containing x . We have $m(J_n) = \delta_n \rightarrow 0$ as $n \rightarrow \infty$. Substituting $J = J_n$ and $f = h$ into the LDT yields:

$$\frac{m_F(J_n)}{m(J_n)} = \frac{\int_{J_n} h \, dm}{m(J_n)} = h_n(x) \rightarrow h(x) \text{ as } n \rightarrow \infty.$$

Since F is absolutely continuous, it places zero mass on singletons, so $F(x) - F(a) = m_F((a, x]) = m_F([a, x])$. The shifted function $G(x) = F(x) - F(a)$ satisfies Corollary 7.35.33 for the finite signed measure m_F , guaranteeing $G'(x) = h(x)$. Since shifting by a constant preserves the derivative, $F'(x) = h(x)$.

As this holds for all $x \notin (D \cup N)$, $F' = h$ m -a.e. Substituting F' for h yields $\frac{dm_F}{dm} = F'$ m -a.e.

Finally, disjoint additivity gives $m_F([a, x]) = m_F((a, x])$. Integrating the Radon-Nikodym derivative over $(a, x]$ yields the Fundamental Theorem of Calculus:

$$F(x) - F(a) = m_F((a, x]) = \int_{(a, x]} F' \, dm = \int_a^x F'(t) \, dt,$$

where the final integral strictly denotes the Lebesgue integral with respect to m . $\square$

Corollary 7.37. If F is absolutely continuous on $[a, b]$ and $F' = 0$ m -a.e., then F is constant.

Proof. Immediate from Theorem 7.36 by substituting $F' = 0$ into $F(x) - F(a) = \int_a^x F'(t) dt = 0$. $\square$

Corollary 7.38. *If $f \in L^1([a, b])$ and $F(x) = \int_a^x f dm$ for each $x \in [a, b]$, then F is differentiable m -a.e. and $F'(x) = f(x)$ for almost every $x \in [a, b]$.*

Proof. By Proposition 7.21, F is absolutely continuous on $[a, b]$. Theorem 7.36 then guarantees F is differentiable m -a.e. Since the Radon-Nikodym derivative is unique up to m -null sets, we immediately conclude $F'(x) = f(x)$ m -a.e. $\square$

Theorem 7.39. *Let ν be a bounded signed measure and let $|\nu|$ be its total variation. Then we can find a measurable function h such that $|h(\omega)| = 1$ for all $\omega \in \Omega$ and $\nu(E) = \int_E h d|\nu|$ for all $E \in \mathcal{F}$.*

CruX. Note $|\nu|$ dominates ν , since $0 \leq |\nu(E)| \leq |\nu|(E)$ for any $E \in \mathcal{F}$. Because ν is bounded, $|\nu|$ is a finite positive measure (Theorem 7.32), guaranteeing $\nu \ll |\nu|$. The Radon-Nikodym Theorem (Corollary 7.33.5) provides a measurable h such that $\nu(E) = \int_E h d|\nu|$.

Let $C_\alpha = \{|h| < \alpha\}$ for some $\alpha \in (0, 1)$. For any partition $\{E_i\}$ of C_α :

$$\sum_{i=1}^{\infty} |\nu(E_i)| = \sum_{i=1}^{\infty} \left| \int_{E_i} h d|\nu| \right| \leq \sum_{i=1}^{\infty} \alpha |\nu|(E_i) = \alpha |\nu|(C_\alpha).$$

Taking the supremum over all partitions on the LHS yields $|\nu|(C_\alpha) \leq \alpha |\nu|(C_\alpha)$. Since $|\nu|$ is finite, $(1 - \alpha) |\nu|(C_\alpha) \leq 0$. Since $\alpha < 1$, C_α is a $|\nu|$ -null set. Hence $|h| \geq 1$ $|\nu|$ -a.e.

Let $D = \{h > 1\}$. If $|\nu|(D) > 0$, strict inequality of the integral (from the note after Proposition 4.17) yields:

$$\nu(D) = \int_D h d|\nu| > \int_D 1 d|\nu| = |\nu|(D).$$

This contradicts the domination property $\nu(D) \leq |\nu|(D)$. Thus, D is $|\nu|$ -null.

Similarly, for $D' = \{h < -1\}$, assuming $|\nu|(D') > 0$ yields:

$$\nu(D') = \int_{D'} h d|\nu| < \int_{D'} -1 d|\nu| = -|\nu|(D').$$

Taking absolute values gives $|\nu(D')| > |\nu|(D')$, again violating domination. Thus, D' is $|\nu|$ -null.

We conclude $|h| \leq 1$ $|\nu|$ -a.e. Combined with $|h| \geq 1$ $|\nu|$ -a.e., we have $|h| = 1$ outside a $|\nu|$ -null set. Redefining $h(\omega) = 1$ on this null set strictly completes the proof. $\square$

Proposition 7.40. *Given a finite measure ρ and a function $f \in L^1(\rho)$, suppose that for every $E \in \mathcal{F}$ with $\rho(E) > 0$, we have*

$$\left| \frac{\int_E f d\rho}{\rho(E)} \right| \leq 1.$$

Then $|f| \leq 1$ ρ -a.e.

Cru. Consider the set $D = \{f > 1\}$. If D is not a ρ -null set ($\rho(D) > 0$), strict integration (from the note after Proposition 4.17) yields:

$$\int_D f \, d\rho > \int_D 1 \, d\rho = \rho(D).$$

This implies $|\int_D f \, d\rho| > \rho(D)$, directly contradicting the hypothesis.

Similarly, let $D' = \{f < -1\}$. If $\rho(D') > 0$, strict integration yields:

$$\int_{D'} f \, d\rho < \int_{D'} -1 \, d\rho = -\rho(D').$$

Taking absolute values gives $|\int_{D'} f \, d\rho| > \rho(D')$, again contradicting the hypothesis.

Therefore, both D and D' must be ρ -null sets, proving $|f| \leq 1$ ρ -a.e. $\square$

Proposition 7.41 (Hahn Decomposition). *Let ν be a bounded signed measure. There are disjoint measurable sets A, B such that $A \cup B = \Omega$ and $\forall F \in \mathcal{F}$, $\nu^+(F) = \nu(B \cap F)$, $\nu^-(F) = -\nu(A \cap F)$. Furthermore, this partition is unique up to $|\nu|$ -null sets. Consequently, if $\nu = \lambda_1 - \lambda_2$ for measures λ_1, λ_2 , then $\lambda_1 \geq \nu^+$ and $\lambda_2 \geq \nu^-$.*

Cru. Theorem 7.39 provides an integrable function h such that $\nu(F) = \int_F h \, d|\nu|$ with $|h(\omega)| = 1$ for all $\omega \in \Omega$. Define $A = \{h = -1\}$ and $B = \{h = 1\}$. Clearly, $A \cup B = \Omega$ and $A \cap B = \emptyset$.

From Definition 7.32.20, $\nu^+ = \frac{1}{2}(|\nu| + \nu)$. Because $\frac{1+h(\omega)}{2} = 0$ for $\omega \in A$ and 1 (which is $h(\omega)$) for $\omega \in B$:

$$\begin{aligned} \nu^+(F) &= \int_F \left(\frac{1+h}{2} \right) d|\nu| \\ &= \int_{F \cap B} 1 \, d|\nu| = \int_{F \cap B} h \, d|\nu| \\ &= \nu(F \cap B). \end{aligned}$$

Similarly, $\nu^- = \frac{1}{2}(|\nu| - \nu)$. Since $\frac{1-h(\omega)}{2} = 1$ (which is $-h(\omega)$) for $\omega \in A$ and 0 for $\omega \in B$:

$$\nu^-(F) = \int_F \left(\frac{1-h}{2} \right) d|\nu| = \int_{F \cap A} -h \, d|\nu| = -\nu(F \cap A).$$

To verify uniqueness up to a $|\nu|$ -null set, let (A', B') be another such partition. Let $C = B' \setminus B$. For any measurable subset $E \subseteq C$, because $E \subseteq B'$, we have $\nu(E) = \nu(E \cap B') = \nu^+(E) \geq 0$. Conversely, because $C \subseteq A$ (since $A = B^c$), $E \subseteq A \implies \nu(E) = -\nu^-(E \cap A) = -\nu^-(E) \leq 0$. Therefore, $\nu(E) = 0$ for all $E \subseteq C$, making C a ν -null set. Symmetrically, $B \setminus B'$ is a ν -null set. Thus, the symmetric difference $B \Delta B'$ is a ν -null set. Since the sets partition Ω , $A \Delta A' = B \Delta B'$, making it a ν -null set (and hence a $|\nu|$ -null set as they are equivalent).

Minimality of this decomposition was established in Proposition 7.32.40. $\square$

Proposition 7.41.14 (Exercise 7.14). *Let ν be a bounded signed measure. For any $F \in \mathcal{F}$, define the set $A_F = \{\nu(G) : G \in \mathcal{F}, G \subseteq F\}$. Then $\nu^+(F) = \sup A_F$ and $\nu^-(F) = -\inf A_F$.*

CruX. Since $\nu = \nu^+ - \nu^-$ and ν^- is positive, $\nu(G) \leq \nu^+(G)$ for any $G \subseteq F$. Because ν^+ is monotonic, $\nu^+(G) \leq \nu^+(F)$. Therefore, $\nu(G) \leq \nu^+(F)$ for all $G \in A_F$, establishing $\nu^+(F)$ as an upper bound.

From the Hahn Decomposition (Proposition 7.41), there exists a positive set B such that $\nu^+(F) = \nu(F \cap B)$. Since $(F \cap B) \in \mathcal{F}$ and $(F \cap B) \subseteq F$, the value $\nu(F \cap B)$ is in A_F .

Because the upper bound is achieved by an element in the set, it is the supremum: $\sup A_F = \nu^+(F)$.

A symmetric argument using the negative set A shows $\nu(G) = \nu^+(G) - \nu^-(G) \geq -\nu^-(G) \geq -\nu^-(F)$. The lower bound is attained at $G = F \cap A$, yielding $\nu^-(F) = -\inf A_F$. $\square$

Proposition 7.41.15 (Exercise 7.15). *Show that when $\nu(F) = \int_F f \, d\mu$ where $f \in L^1(\mu)$ for a positive measure μ , the Hahn decomposition sets are $A = \{f < 0\}$ and $B = \{f \geq 0\}$, and $\nu^+(F) = \int_F f^+ \, d\mu$, while $\nu^-(F) = \int_F f^- \, d\mu$.*

CruX. Let $A = \{f < 0\}$ and $B = \{f \geq 0\}$. For any partition $\{F_i\}$ of F , the triangle inequality gives:

$$\sum_{i=1}^{\infty} |\nu(F_i)| = \sum_{i=1}^{\infty} \left| \int_{F_i} f \, d\mu \right| \leq \sum_{i=1}^{\infty} \int_{F_i} |f| \, d\mu = \int_F |f| \, d\mu.$$

This shows $\int_F |f| \, d\mu$ is an upper bound for the total variation.

Construct a specific partition using A and B : let $F_1 = F \cap B$ and $F_2 = F \cap A$. Then:

$$|\nu(F_1)| + |\nu(F_2)| = \left| \int_{F \cap B} f \, d\mu \right| + \left| \int_{F \cap A} f \, d\mu \right|.$$

Because $f \geq 0$ on B and $f < 0$ on A , the absolute values act perfectly on the integrands:

$$\int_{F \cap B} f \, d\mu - \int_{F \cap A} f \, d\mu = \int_{F \cap B} |f| \, d\mu + \int_{F \cap A} |f| \, d\mu = \int_F |f| \, d\mu.$$

Since this partition achieves the supremum, $|\nu|(F) = \int_F |f| \, d\mu$.

Applying the algebraic definitions of ν^+ and ν^- :

$$\begin{aligned} \nu^+(F) &= \frac{1}{2} \int_F (|f| + f) \, d\mu = \int_F f^+ \, d\mu \\ \nu^-(F) &= \frac{1}{2} \int_F (|f| - f) \, d\mu = \int_F f^- \, d\mu \end{aligned}$$

By definition, $f^+ = f$ on B and 0 otherwise. Similarly, $f^- = -f$ on A and 0 otherwise. Thus:

$$\begin{aligned}\nu^+(F) &= \int_{F \cap B} f \, d\mu = \nu(F \cap B) \\ \nu^-(F) &= \int_{F \cap A} -f \, d\mu = -\nu(F \cap A)\end{aligned}$$

Because $A \cup B = \Omega$ and $A \cap B = \emptyset$, $A = \{f < 0\}$ and $B = \{f \geq 0\}$ satisfy the Hahn decomposition conditions perfectly. By Proposition 7.41, these are unique up to $|\nu|$ -null sets. $\square$

Definition 7.42. Let μ be a signed measure and f a measurable function on $F \in \mathcal{F}$. Define the integral $\int_F f \, d\mu$ by

$$\int_F f \, d\mu = \int_F f \, d\mu^+ - \int_F f \, d\mu^-$$

whenever both terms on the right are finite or not of the form $\pm(\infty - \infty)$. The function is *summable* if the integral so defined is finite.

Proposition 7.42.16 (Exercise 7.16). Let μ be a finite measure and define the signed measure ν by $\nu(F) = \int_F g \, d\mu$. Then $f \in L^1(\nu)$ if and only if $fg \in L^1(\mu)$ and $\int_E f \, d\nu = \int_E fg \, d\mu$ for all μ -measurable sets E .

Note: We assume $g \in L^1(\mu)$; otherwise, ν would not be a bounded signed measure.

CruX. Since $g \in L^1(\mu)$, ν is a bounded signed measure, and its total variation $|\nu|$ is a finite positive measure. By Proposition 7.41.15, $|\nu|(F) = \int_F |g| \, d\mu$, with variations $\nu^+(F) = \int_F g^+ \, d\mu$ and $\nu^-(F) = \int_F g^- \, d\mu$.

By definition, $f \in L^1(\nu) \iff \int_\Omega |f| \, d|\nu| < \infty$. Since μ and $|\nu|$ are finite positive measures, the change-of-measure formula (Proposition 7.6) with $d|\nu| = |g| \, d\mu$ yields:

$$\int_\Omega |f| \, d|\nu| = \int_\Omega |f| |g| \, d\mu = \int_\Omega |fg| \, d\mu.$$

Thus, $\int_\Omega |f| \, d|\nu| < \infty \iff \int_\Omega |fg| \, d\mu < \infty$, strictly proving $f \in L^1(\nu) \iff fg \in L^1(\mu)$.

To prove equality over E , substitute the positive and negative variations into Definition 7.42:

$$\begin{aligned}\int_E f \, d\nu &= \int_E f \, d\nu^+ - \int_E f \, d\nu^- \\ &= \int_E fg^+ \, d\mu - \int_E fg^- \, d\mu \\ &= \int_E f(g^+ - g^-) \, d\mu \\ &= \int_E fg \, d\mu.\end{aligned}$$

(Since $f \in L^1(\nu)$, it is strictly integrable w.r.t. both ν^+ and ν^- , making the change-of-measure steps rigorously valid.) $\square$

7.4 Probability

In Theorem 5.43, a functional analytic construction established that for any $X \in L^1(\mathcal{F})$, the conditional expectation $Y = \mathbb{E}(X|\mathcal{G})$ exists and satisfies

$$\int_G Y \, dP = \int_G X \, dP \text{ for all } G \in \mathcal{G} \quad (7.4)$$

where $\mathcal{G}$ is a sub- σ -field of $\mathcal{F}$. The existence of this conditional expectation is trivialized by the Radon-Nikodym theorem. Let $\mu(F) \triangleq \int_F X \, dP$. Let ν and Q be restrictions of μ and P to $\mathcal{G}$. If $Q(N) = 0$ for $N \in \mathcal{G}$, then $P(N) = 0 \implies \mu(N) = 0 \implies \nu(N) = 0$. Hence, $\nu \ll Q$. The Radon-Nikodym Theorem (Corollary 7.33.5) guarantees a $Y \in L^1(\mathcal{G})$ such that $\nu(G) = \int_G Y \, dQ$ for all $G \in \mathcal{G}$. Since $\nu(G) = \mu(G)$, we have $\int_G X \, dP = \int_G Y \, dP$, establishing existence.

In particular, for $X \in L^2(\mathcal{F})$, its orthogonal projection onto $L^2(\mathcal{G})$ is a version of the Radon-Nikodym derivative of $\mu : F \mapsto \int_F X \, dP$.

Note: Let Y be $\mathcal{G}$ -measurable and X be $\mathcal{F}$ -measurable, where $\mathcal{G} \subseteq \mathcal{F}$.

- $\int_G X \, dP$: Well-defined for any $G \in \mathcal{F}$.
- $\int_G Y \, dP$: Well-defined for any $G \in \mathcal{F}$, because Y is automatically $\mathcal{F}$ -measurable.
- $\int_G Y \, dQ$: Well-defined only if $G \in \mathcal{G}$. If $Q = P|_{\mathcal{G}}$, $\int_G Y \, dQ = \int_G Y \, dP$.
- $\int_G X \, dQ$: Undefined. X is not generally $\mathcal{G}$ -measurable, so Q cannot measure its pre-images.

Definition 7.44. A random variable $\mathbb{E}(X|\mathcal{G})$ is the conditional expectation of X relative to $\mathcal{G}$ if:

- (i) $\mathbb{E}(X|\mathcal{G})$ is $\mathcal{G}$ -measurable.
- (ii) $\int_G \mathbb{E}(X|\mathcal{G}) \, dP = \int_G X \, dP$ for all $G \in \mathcal{G}$.

Conventions: $X, Y, X_n \in L^1(\Omega, \mathcal{F}, P)$; $\mathcal{G}, \mathcal{H}$ are sub- σ -fields of $\mathcal{F}$.

Proposition 7.45. *Properties of $\mathbb{E}(X|\mathcal{G})$:*

- (i) $\mathbb{E}(\mathbb{E}(X|\mathcal{G})) = \mathbb{E}(X)$.
- (ii) If X is $\mathcal{G}$ -measurable, then $\mathbb{E}(X|\mathcal{G}) = X$.
- (iii) If X is independent of $\mathcal{G}$, then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$.
- (iv) *Linearity:* $\mathbb{E}((aX + bY)|\mathcal{G}) = a\mathbb{E}(X|\mathcal{G}) + b\mathbb{E}(Y|\mathcal{G})$.

CruX. (i) Set $G = \Omega$ in Def 7.44(ii): $\int_{\Omega} \mathbb{E}(X|\mathcal{G}) \, dP = \int_{\Omega} X \, dP = \mathbb{E}(X)$.

- (ii) X trivially satisfies both Def 7.44(i) and (ii). By P -a.e. uniqueness of the Radon-Nikodym derivative, $\mathbb{E}(X|\mathcal{G}) = X$.
- (iii) If X is independent of $\mathcal{G}$, then X and $\mathbb{I}_G$ are independent for any $G \in \mathcal{G}$. By Theorem 5.36, $\int_G X \, dP = \mathbb{E}(X\mathbb{I}_G) = \mathbb{E}(X)P(G) = \int_G \mathbb{E}(X) \, dP$. Since the constant $\mathbb{E}(X)$ is $\mathcal{G}$ -measurable, $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$ P -a.e.
- (iv) Follows immediately from the linearity of the Lebesgue integral.

□

Theorem 7.46. *Properties of conditional expectation:*

- (i) If $X \geq 0$, then $\mathbb{E}(X|\mathcal{G}) \geq 0$ P -a.s.
- (ii) If $X_n \geq 0$ and $X_n \uparrow X$ a.s., then $\mathbb{E}(X_n|\mathcal{G}) \uparrow \mathbb{E}(X|\mathcal{G})$ a.s.
- (iii) If Y is $\mathcal{G}$ -measurable and $XY \in L^1$, then $\mathbb{E}(XY|\mathcal{G}) = Y\mathbb{E}(X|\mathcal{G})$.
- (iv) If $Y = \mathbb{E}(X|\mathcal{G})$ and $\mathcal{H} \subseteq \mathcal{G}$, then $\mathbb{E}(Y|\mathcal{H}) = \mathbb{E}(X|\mathcal{H})$.
- (v) (Conditional Jensen) If φ is convex and $\varphi(X) \in L^1(P)$, then $\mathbb{E}(\varphi(X)|\mathcal{G}) \geq \varphi(\mathbb{E}(X|\mathcal{G}))$.

Crux. (i) If $X \geq 0$, then $\int_G X \, dP \geq 0 \implies \int_G \mathbb{E}(X|\mathcal{G}) \, dP \geq 0$ for all $G \in \mathcal{G}$. By Prop 4.22(i), $\mathbb{E}(X|\mathcal{G}) \geq 0$ P -a.s.

- (ii) Since $X_{n+1} - X_n \geq 0$, (i) and linearity imply $Y_n = \mathbb{E}(X_n|\mathcal{G})$ is a.s. increasing to some $\mathcal{G}$ -measurable Z . By the Monotone Convergence Theorem, for any $G \in \mathcal{G}$:

$$\int_G Z \, dP = \lim_n \int_G Y_n \, dP = \lim_n \int_G X_n \, dP = \int_G X \, dP.$$

Thus $Z = \mathbb{E}(X|\mathcal{G})$ a.s.

- (iii) For $Y = \mathbb{I}_{G'}$ ($G' \in \mathcal{G}$): $\int_G \mathbb{I}_{G'} X \, dP = \int_{G \cap G'} X \, dP = \int_{G \cap G'} \mathbb{E}(X|\mathcal{G}) \, dP = \int_G \mathbb{I}_{G'} \mathbb{E}(X|\mathcal{G}) \, dP$. Linearity extends this to simple functions. The MCT extends it to non-negative Y , and decomposition into Y^+, Y^- completes it for general Y .
- (iv) Let $H \in \mathcal{H}$. By definition, $\int_H X \, dP = \int_H \mathbb{E}(X|\mathcal{H}) \, dP$. Since $H \in \mathcal{G}$, we also have $\int_H X \, dP = \int_H Y \, dP = \int_H \mathbb{E}(Y|\mathcal{H}) \, dP$. Since both bounds are $\mathcal{H}$ -measurable, they are equal P -a.s.
- (v) We take a few results from basic real analysis about convex functions as given (see Appendix):

- (1) The left- and right-hand derivatives of a convex function exist everywhere.
- (2) If we denote $D^+(z)$ and $D^-(z)$ as the right- and left-hand derivatives of φ at z , then for any $x < z$, $\frac{\varphi(x) - \varphi(z)}{x - z} \leq D^-(z)$, and for any $x > z$, $D^+(z) \leq \frac{\varphi(x) - \varphi(z)}{x - z}$.
- (3) The function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is locally Lipschitz, which implies it is continuous everywhere.

With this, we define the function

$$c(z) \triangleq \frac{D^+(z) + D^-(z)}{2}$$

which equals the derivative of φ wherever it is differentiable. Since $D^-(z) \leq c(z) \leq D^+(z)$, applying Fact (2) and multiplying by $(x - z)$ (which flips the inequality when $x < z$) guarantees that for all x :

$$\varphi(x) \geq \varphi(z) + c(z)(x - z).$$

Let $Z = \mathbb{E}(X|\mathcal{G})$. Substituting X and Z , we note $c(Z)$ (monotonic) and $\varphi(X)$ (continuous) are $\mathcal{G}$ -measurable and $\mathcal{F}$ -measurable respectively. Taking conditional expectations yields:

$$\mathbb{E}(\varphi(X)|\mathcal{G}) \geq \mathbb{E}(\varphi(Z)|\mathcal{G}) + \mathbb{E}(c(Z)(X - Z)|\mathcal{G}).$$

Since Z is $\mathcal{G}$ -measurable and φ is continuous, $\varphi(Z)$ is $\mathcal{G}$ -measurable, so $\mathbb{E}(\varphi(Z)|\mathcal{G}) = \varphi(Z)$. Since $c(Z)$ is $\mathcal{G}$ -measurable, Part (iii) pulls it out:

$$\mathbb{E}(c(Z)(X - Z)|\mathcal{G}) = c(Z)\mathbb{E}(X - Z|\mathcal{G}) = c(Z)(Z - Z) = 0.$$

Thus, $\mathbb{E}(\varphi(X)|\mathcal{G}) \geq \varphi(\mathbb{E}(X|\mathcal{G}))$.

□

Setting $\varphi(x) = |x|^p$ for $p \geq 1$ in Theorem 7.46(v) directly shows the L^p norm of the conditional expectation is bounded by the norm of the random variable: $\|\mathbb{E}(X|\mathcal{G})\|_p \leq \|X\|_p$.